



Pharmacology Workbook



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INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.



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AUTONOMIC NERVOUS SYSTEM PHARMACOLOGY

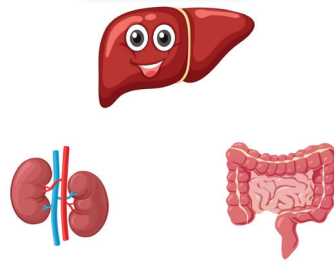


Active space

MIST

Pharmaco-Kinetics

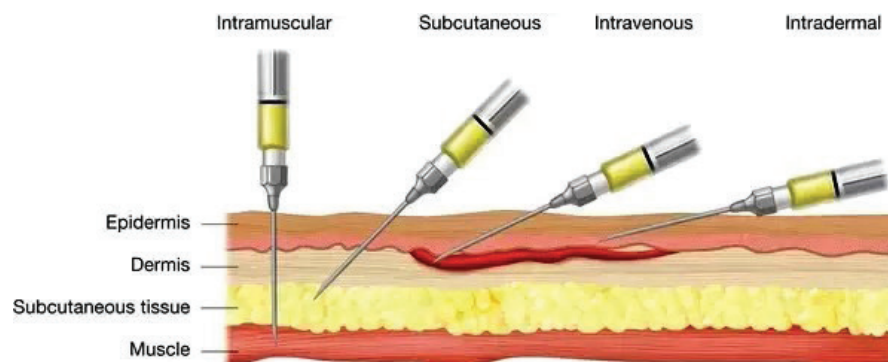
Pharmaco-dynamics

AbsorptionDistributionMetabolismExcretion**ROUTES OF ADMINISTRATION (SYSTEMIC)**

◆ Enteral (Oral, Rectal)

- **Oral** - Most drugs except tide/ase which are peptides broken down in GIT never given by oral route
- **Rectal** - Diazepam

◆ Parenteral

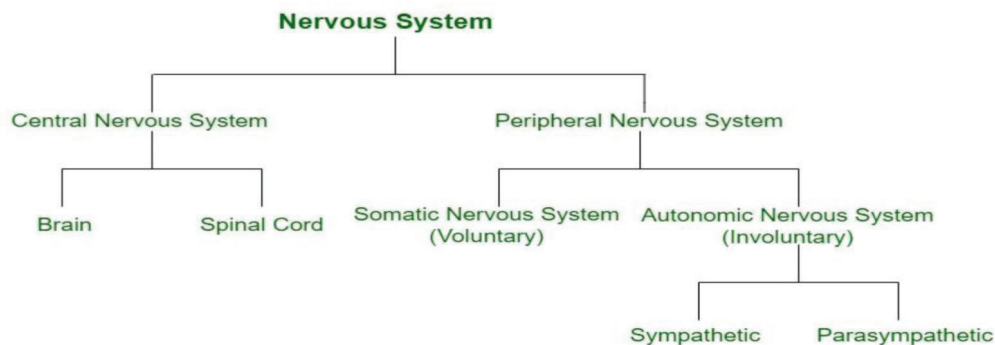


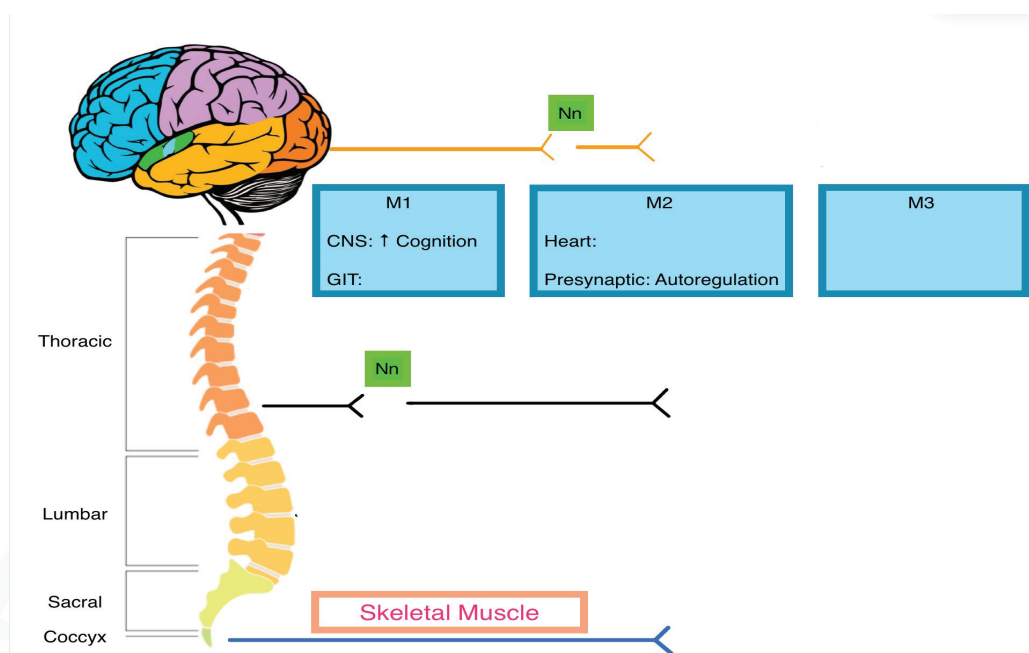
a. Injectable

- **I/M** -
- **S/C** -, Contraceptives
- **I/V** - Nicardipine, Furosemide, Lorazepam
- **Intradermal** - TB & Allergy Testing, BCG

b. Non injectable

- Sublingual - NTG, IDN, Nifedipine
- Intranasal - Desmopressin, GnRH agonists (Nafarelin), Calcitonin
- Inhalational - Insulin, SABA, Anticholinergics in COPD, ICS, NO
- Transdermal - NTG, Scopalamine, Nicotine, Diphenhydramine, Rivastigmine, Selegiline, Estrogen



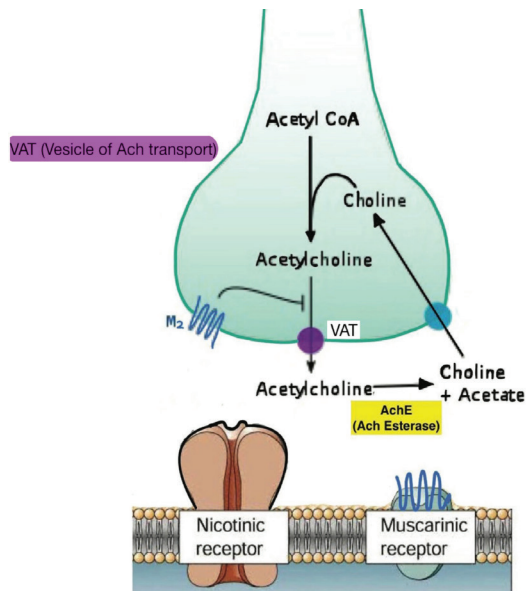


Parasympathetic/Cholinergic	Sympathetic/Adrenergic
<u>D</u> efecation	Defecation ↓
<u>U</u> rination	Urination ↓
<u>M</u> iosis+Accommodation (ciliary muscle) (iris sphincter)	Mydriasis (iris dilator)
<u>B</u>	Bronchodilation
<u>B</u> radycardia (d/t delay in depolarization)	Tachycardia
<u>E</u>	
<u>L</u>	
<u>S</u>	

PARASYMPATHETIC NERVOUS SYSTEM

Ach: works on Nicotinic and Muscarinic receptors

It can be used topically in ophthalmic Sx but doesn't have systemic use



PARASYMPATHOMIMETICS / CHOLINERGIC AGONIST

✦ Direct agonists

Tertiary amines	Quarternary amines
Pilocarpine: DOC in CAG Also used in Xerostomia SE: Brow ache, Retinal detachment Cevimeline	Bethenechol:

Active space

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Carbachol:

Methacholine:

Indirect agonists (Carbamates-Reversible)

Tertiary amines	Quarternary amines
Non-carbamates Donepezil (DOC) Gallantamine Rivastigmine Tacrine Physostigmine	 Suggamadex: Rx of Rocuronium & Vecuronium toxicity



Datura aka Jimsonweed/
Jamestown was previously used
in Asthma Rx

Atropine overdose: Hot as a
Hare (Hyperthermia), Blind as a
bat (Mydriasis), Red as a beet
(Flushed face), Dry as a bone
(↓ Secretions), Mad as a hatter
(Delirium)

Edrophonium: Helps differentiate
Myasthenia (symptoms improves)
vs Cholinergic crisis (symptoms
worsens)

Aminopyridines (K-channel) increases activation of voltage gated
Ca channels releasing Ach causing muscle contraction hence used in
Lambert-eaton syndrome

Organophosphates (Irreversible) Carbamates (Reversible)

Ex of OP: Parathion, Malathion, Chlorpyrifos, Nerve gas (Tabun, Sarin,
Soman), Carbamates (Carbaryl, Propoxur)

Endrin is an organochloride for which no antidote is available

MOA:

C/F:

- + M: Diarrhea, Urination, Miosis, B.C, Bradycardia, Emesis,
Lacrimation, Salivation
- + Nm: Initially contractions later SMR

OP: bind to esteric site only whereas Carbamate binds to both anionic & esteric site

Garlicky odour could be noted

Rx:

i.v Atropine (DOC): every 5 minutes until "Signs of Atropinization (Mydriasis, ↑H.R, ↓Secretions) seen

In OP poisoning AchE reactivators (oximes) are added

Its also DOC for early mushroom (inocybe species) poisoning

Late mushroom poisoning (Amanita phalloides) causes hepatic failure), there is no Rx

PARASYMPATHOLYTICS / ANTI-CHOLINERGICS

- Vasamicol : ⊖ VAT (Vesicle of ACH transport)
- Hemicholinium :
- Botox: inhibits release of Ach by acting on SNARE (help in fusion of vesicle & exocytosis of NT)

Uses: in Rx of Blepharospasm, Wrinkles, Glabellar lines, Cervical dystonia, Achalasia cardia

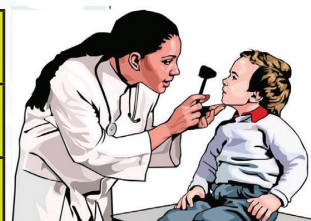
Onabotulinum toxin A: Axillary hyperhidrosis, Prophylaxis of Migraine

MUSCARINIC ANTAGONISTS

Defecation () Urination () Mydriasis, B.D, H.R ()

- ◆ **Bronchi:** SAMA- Ipratropium, IAMA- Oxitropium, LAMA- Revefenacin, Umeclidinium, Tiotropium SE: Dry mouth (MC)
They don't interfere with muco-ciliary clearance unlike atropine that inhibits it and cause accumulation of secretions and mucus plugs
- ◆ **Anti-secretory:** (Quarternary amine): Glycopyrrolate
- ◆ **Heart:** Atropine - DOC for bradycardia (0.5-1.0 mg I.V), AV block
- ◆ **Eyes:**

Cyclopentolate (1%)	Mod Cyclo	12-24 hrs.	Older children, adult
Atropine (1%)	Strong Cyclo	7-24 days	Children
Tropicamide (0.5%)	Wear Cyclo Fastest	4-6 hrs.	Adult



They can precipitate glaucoma

- ◆ **CNS:** Benzhexol (Trihexyphenidyl)
- ◆ **Urinary system:** Rx for overactive bladder (Detrusor instability, urge incontinence)

Solifenacin (M3 selective)

Oxybutynin (Direct smooth muscle relaxant)

Fesoterodine, Flavoxate (Direct smooth muscle muscle relaxant)

Tolterodine, Trospium (Don't Cross BBB)

Detrusor: Darifenacin (M3 selective)

Solifenacin, Tolterodine Darifenacin (STD) are uroselective

Active space

MIST

✦ **GIT:**

Specific M1 $\ominus$: Piren/Telen

Anti-spasmodics: Dicyclomine (also has direct anti-spasmodic effect)

Anti-emetic: Scopolamine (Hyoscine) Px of Motion sickness

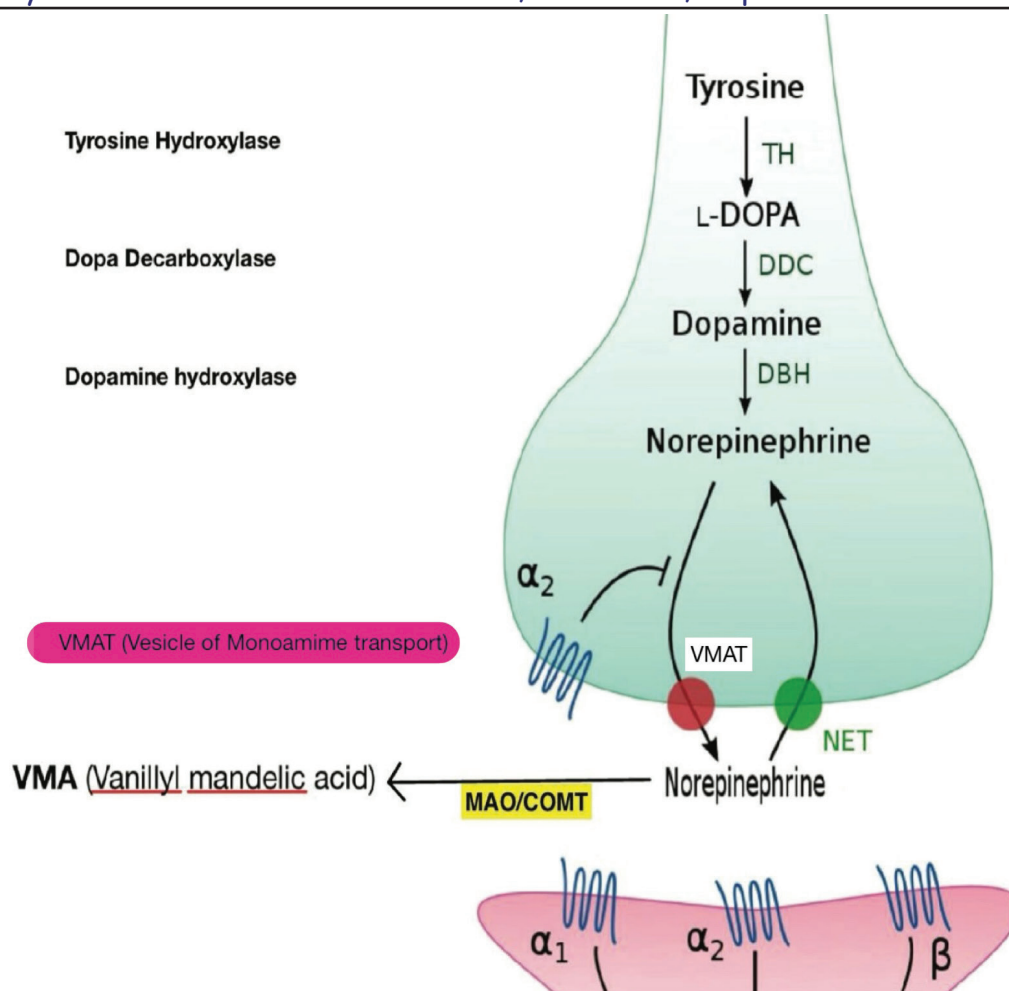
- ✓ Transdermal patch (behind ear) → take 6 hrs before journey (works for 3 days)
 - ✓ Tablets → take 30 mins - 1hr before journey (works upto 6 hrs)
- Also used in Narcoanalysis (DOC: Thiopentone aka Pentothal Sodium)

SYMPATHETIC NERVOUS SYSTEM

α_1	α_2
α_1a : mydriasis, urethra constriction (prostate) α_1b : vasoconstriction	α_2a : ↓ symp func, ↓ functions in CNS α_2b (postsynaptic): V.C

β_1	β_2	β_3
↑ Dromotropy ↑ Chronotropy ↑ Inotropy ↑ renin production	B.D, V.D, Uterine relax ↑ glycemia ↓ kalemia Tremors	↓ Urination Lipolysis

Synthesis of Catecholamines : Adr, Nor - Adr, Dop



SYMPATHOMIMETICS

Catecholamines

- ✓ Adrenaline (Epinephrine):
- ✓ Nor-adrenaline (Nor-epinephrine):
- ✓ Dopamine:
- ✓ Dobutamine: ⊕B1 >>> ⊕B2
- ✓ Isoprenaline: ⊕B1, ⊕B2
- ✓ Fenoldopam: ⊕D1
- ✓ Dopexamine: ⊕D1, ⊕B2

◆ Adrenaline:

- DOC: shock (I.m> S.C) 0.5 ml of 1:1000 if NR → 0.5-1ml iv (1:10000)
- Cardiac arrest: i.v > intraosseous (1:10000), Bradycardia
- Bronchial asthma 0.5 ml of 1:1000 (i.m/s.c)
- Given along with Local anesthetics (↑ Duration of caine)
- Glaucoma (prodrug: dipivefrine) SE - Conjunctival Pigmentation
- Shows biphasic response on BP (High BP due to α_1 followed by Low BP due to β_2)
- Shows Vasomotor reversal of Dale/ Epinephrine reversal
- Re-reversal is seen with Adr + Beta blockers

Adr only		Nor-Adr only	Dale Experiment Add $\alpha \ominus$	
$\oplus \alpha_1$	$\oplus \beta_2$	$\oplus \alpha_1$	Adr + $\alpha \ominus$	Nor-adr + $\alpha \ominus$
			$\oplus \alpha_1$ $\oplus \beta_2$	$\oplus \alpha_1$
B.P:		B.P:	B.P:	B.P:

Dilutions of epinephrine

- 1: 1000: Im or subcutaneous or endotracheal route.
- 1: 10,000: IV or intraosseous or intracardiac route.
- 1:1,00,000/2,00,000: Along with local anaesthetic.

Nor-Adrenaline: DOC in vasodilatory shock & cardiogenic shock (with hypotension)

✓ **Dopamine:** DOC for Cardiogenic shock with Oliguria, Acute CHF with Oliguria

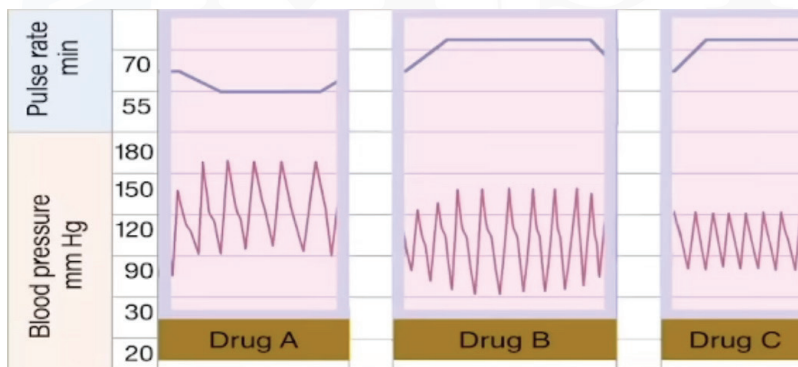
Low dose ($<2 \mu\text{g}/\text{kg}/\text{min}$):

Med dose ($2-10 \mu\text{g}/\text{kg}/\text{min}$):

High dose ($10-15 \mu\text{g}/\text{kg}/\text{min}$):

	Adrenaline- α_1 , β_1 , β_2	Nor-adrenaline- α_1 , β_1	Isoprenaline- β_1 , β_2
Mean BP	↑	↑↑	↓
Heart rate	Direct: ↑↑ Indirect: ↓	Direct: ↑ Indirect: ↓↓	Direct: ↑ Indirect: ↑
Net effect (HR)	↑	↓	↑↑

Nor-Adr ↑ H.R in transplanted heart or patient on atropine



Dobutamine: DOC-Inotrope in Acute CHF, Cardiogenic shock with ↓ EF
also used in stress echocardiography

Isoprenaline / Isoproterenol:

Fenoldopam:

INDIRECTLY ACTING SYMPATHOMIMETICS

◆ Reuptake $\ominus$: Cocaine

All local anesthetics are v.d except Cocaine (v.c).

MOA of local anesthetic $\ominus$ Na

Antidote of last (Local anesthetics systemictox) : 20% Intralipid
20% Intralipid acts mainly by the "lipid sink" mechanism-
sequestering lipophilic local anesthetic molecules from tissues
(heart, brain) into an intravascular lipid phase

◆ Release enhancers or drugs causing displacement of NT:

- ✓ Ephedrine: Nasal decongestant, Rx hypotension d/t spinal anesthesia with bradycardia
- ✓ Amphetamines:
- ✓ Tyramine: Cheese Rx (Severe HTN)

Tachyphylaxis (rapid tolerance) seen with most of these drugs

- ✓ Methylphenidate:
- ✓ Modafinil:

Tiprolisant (H3 blocker-inverse agonist): Rx of Narcolepsy

NON-CATECHOLAMINES

◆ Alpha-1 agonist:

- Phenylephrine : Active Mydiatric (without cycloplegia),
Decongestant, DOC for Hypotension d/t Spinal anesthesia,
Priapism. SE: Bradycardia

- ❑ Methoxamine, Mephenteramine, Midodrine (DOC)
- ❑ Napha/Oxymeta/Xylometa
SE: Rhinitis medicamentosa/Rebound rhinitis
- ◆ Alpha-2 agonist:
- ❑ Methyl dopa
SE: Coomb's test +ve (check for hemolysis if present discontinue drug)
- ❑ Clonidine: DOC in HTN urgency, used in clonidine suppression test for Dx of Pheochromocytoma, Tics (Tourettes)
SE: Rebound Hypertension
- ❑ Apraclonidine (SE: Lid retraction), Brimonidine (SE: Respiratory depression) used in Rx of Glaucoma
- ❑ Dexmedetomidine: used as pre-anesthetic medication (sedation, ↓ secretions etc)
- ❑ Tizanidine: Used as a central acting muscle relaxant
- ◆ Beta-3 agonist:
Mira / Vibegron : DOC for Urinary incontinence
- ◆ Beta-2 agonist:
- ❑ Bronchodilators - SABA(<4 hr)-Salbutamol, Terbutaline, LABA (12 hr)-Sal & Formoterol
SE: Tremors (d/t direct action on muscles and hypokalemia), Tachycardia (Palpitations), Hypokalemia, Hyperglycemia
- Tocolytic - Ritodrine, Isoxsuprine



SYMPATHOLYTICS

♦ Alpha blockers:

◆ **Non-selective:** **Irreversible:** Phenoxybenzamine

Reversible: Phentolamine, Tolazoline

◆ **Alpha-1 blockers:** Pra / Tera **zosin**

SE: 1st dose hypotension, Postural hypotension

◆ **Selective Alpha-1a blockers:** Tamsulosin, Sildenafil
DOC for Rx of BPH

Tamsulosin is also used for Rx of post-op urinary retention in
bronchospastic disorders

SE: Retrograde ejaculation (d/t urethral sphincter dilation during
ejaculation)

Alpha-2 antagonist: Yohimbine - previously used for erectile dysfunction

♦ Beta blockers:

◆ 1st gen (Non-selective): Propranolol, Timolol, Nadolol, Pindolol

◆ 2nd gen (Cardioselective): Atenolol, Acebutalol, Metoprolol,
Betaxolol, Bisoprolol, Esmolol (metabolized by Esterases)

◆ 3rd gen: additional properties like

- ⇒ **Alpha $\ominus$ + Non-**Sel B $\ominus$** : Labetolol, Carvedilol, Bucindolol,**
- ⇒ **CCB** : Carvedilol - has anti-oxidant property
- ⇒ **NO release** : Nebivolol, Nipradilol (Most cardiosel with anti - oxidant property)
- ⇒ **K channel open** : Tilisolol, **K channel close** Sotalol
- ⇒ **B₂ agonist** : Celiprolol,

Partial agonist/ intrinsic sympathomimetic activity - safe in bradycardia

Can: Celiprolol **Partially:** Pindolol **Agonise:** Acebutalol

Membrane Stabilising Property (Local anesthetic/Na $\ominus$ activity) - C/I in glaucoma d/t anesthetic effect that can inhibit corneal reflex

Posess: Propranolol (Max), **Membrane Stabilising:** Metoprolol **Local:** Labetolol **Anesthetic:** Acebutalol **Property:** Pindolol

Water solubility: Dont cross BBB so no CNS side effects.
C/I in Renal failure

C: Celiprolol **N:** Nadolol **S:** Sotalol **Absorption:** Atenolol

Carvedilol, Metoprolol, Labetolol are safe in renal failure

Uses:

C/I of Beta blockers:

Very:

B:

A:

D:

Choice:

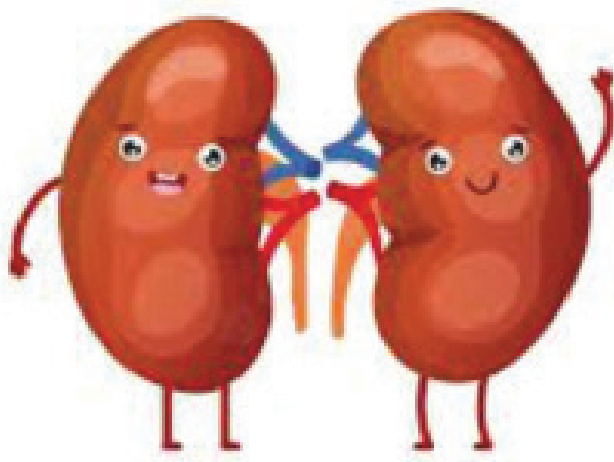
2nd Gen are safe in Asthma, DM and PVD

Antidote: Glucagon (↑ C-AMP that was decreased by B ⊖)

Active space



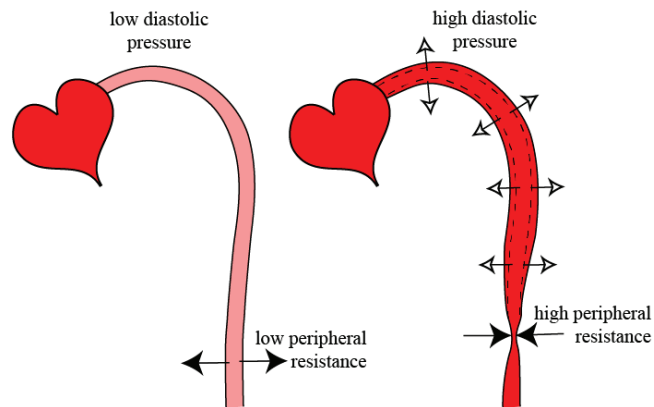
CARDIOVASCULAR AND RENAL PHARMACOLOGY



Active space

ANTI HYPERTENSIVES

AIM : ↓ Total peripheral resistance (TPR)/Systemic vascular resistance (SVR)/Peripheral vascular resistance (PVR)



Drugs altering sympathetic activity:

♦ α₂ agonist - α-methyldopa, clo/moxo/rilme **nidine**
(nidine have extra imidazoline - I₁ receptor agonist)

♦ α₁ blockers - Pra / Tera / Doxa **zosin**

↓ arteriolar and venous resistance with.....effect on lipid profile

SE: 1st dose hypotension, Orthostatic hypotension, Urinary incontinence

♦ β blockers - "olol" ↓ renin, ↓ HR

SE: Cardiovascular depression, Fatigue, Sexual dysfunction, ↑ LDL & TG

Cautions in use: Asthma, Vasospastic diseases, Diabetes

Direct acting vasodilators:♦ Drugs acting through NO

- ⇒ Hydralazine - ↓ TPR via arteriolar dilation. SE -
- ⇒ Nitroprusside - ↓ TPR via arteriolar & venous dilation SE -
..... tox, Hypothyroidism

♦ K channel openers

- ⇒ Minoxidil, Diazoxide, Hydralazine
- ⇒ Topical Minoxidil is used in
- ⇒ Diazoxide is used in

♦ CCB: Block L type Ca channels in heart and blood vessels

Dihydropyridines (DHP) : Amlo/Beni/clini

Non - DHP : Verapamil, diltiazem → Angina, Antiarrhythmics

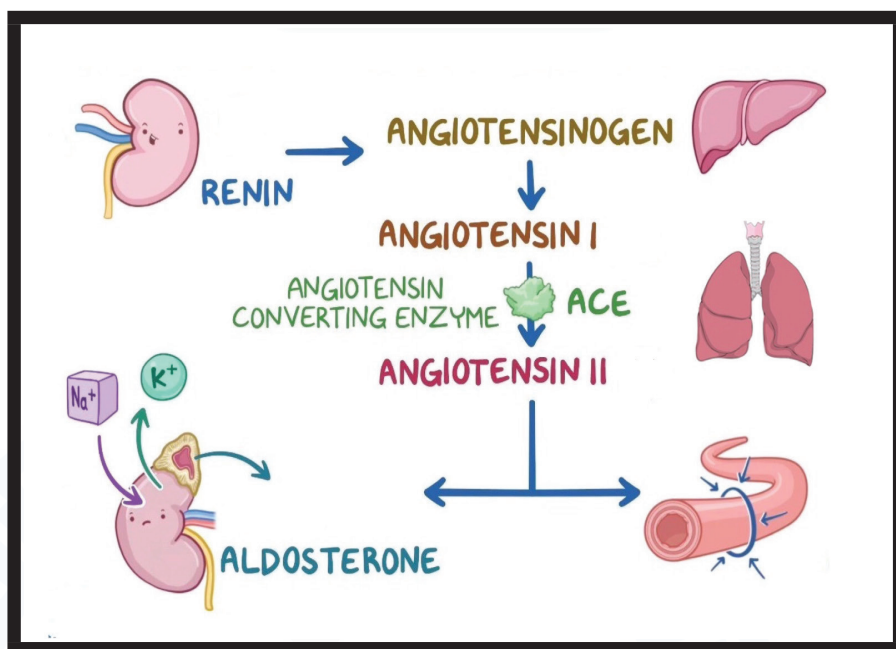
- Nicardipine
- Nifedipine
- Nimodipine is cerebroselective so its used to reverse compensatory v.c (ischemia) after Sub-Arochnoid H'ge
- Clevidipine (Shortest acting as its metabolised by plasma esterase), i.v used in acute HTN episode
- Side effects: Reflex tachycardia (dipines), Gingival hyperplasia Constipation (verapamil), Ankle edema

Gingival Hyperplasia

Phenytoin

CCB

Cyclosporine



All RAAS inhibitors can cause hyperkalemia

- ◆ **Beta-blockers:**
- ◆ **Direct Renin Inhibitor:** Alis / Enal
- ◆ **ACEI:** Capto(shortest)/Lisino / Perindo(longest acting)
- ◆ **ARB's:** Epro (shortest) / Telmi (longest acting)
- All are prodrugs except capto & lisinopril, olme & candesartan
- Uses: HTN, HTN emergency (Enalaprilat - active form of enalapril), Diabetic nephropathy, CKD, MI, Heart failure
- Losartan has anti-platelet, uricosuric
Telmisartan has PPAR-γ agonist activity

SE: ACEI cause (MC),

- ACEI/ARB'S : C/I in pregnancy (Renal agenesis) & bilateral renal artery stenosis



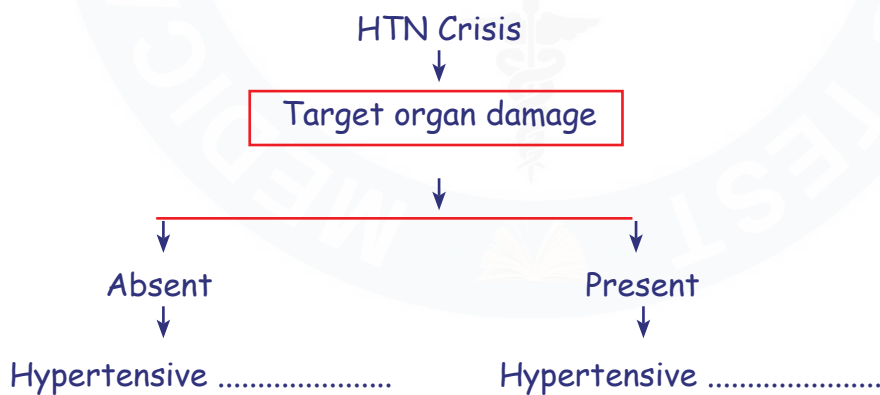
- ✓ **Aldosterone (-)** :
- ✓ Spironolactone: DOC in Rx of Resistant HTN
Lorundrostat - recently approved for resistant HTN
- ✓ **Endothelin receptor (-)** : Aprositentan

Anti HTN Drugs Safe in Pregnancy		
Better	→	$\beta\ominus$ [Labetalol]
Mother	→	Methyldopa
Care	→	Clonidine
During	→	DHP (CCB)
Hypertensive	→	Hydrallazine
Pregnancy	→	Prazosin $\ominus$

Hereditary angioedema:

B2 (Bradykinin) receptor antagonist: Icatibant (s.c)

Kallikrein inhibitor: Ecallantide (s.c), Berostralstat (oral)



♦ HTN with comorbidities :

DM / CKD/Scleroderma	
Angina/ Hyperthyroidism/Migraine/ Anxiety/ Tremors	
Osteoporosis	
Heart failure	
BPH, dyslipidemia	
Raynauds disease	

♦ Pulmonary HTN: (Mnemonic: **Grilled Chicken, PEP SI**)

Guanylate cyclase activators: Rio/Cinaciguat

Calcium channel blockers: Amlodipine, Diltiazem

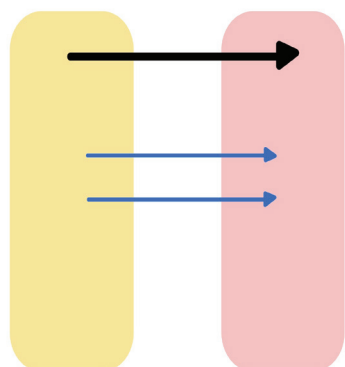
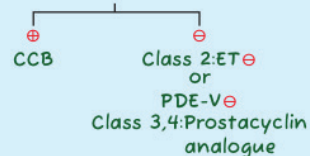
Prostacyclins receptor agonist: Selexipag,

ET1 (ETA) $\ominus$: Ambre/Bo **sentan**

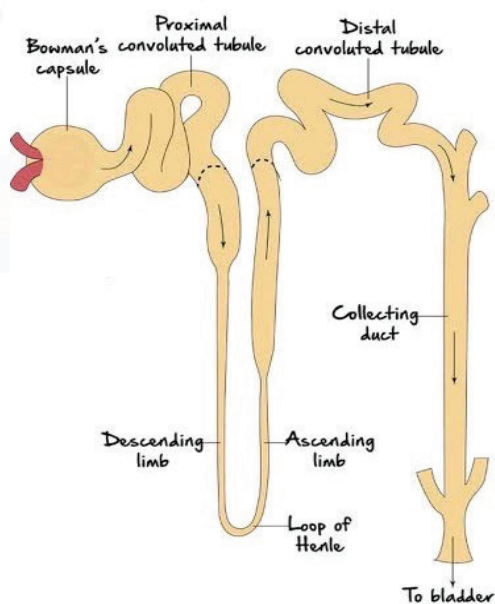
PGI2/Prostacyclin analogue: Epoprostenol, Ileoprost

Sildenafil (Viagra), Tadalafil (long acting) : PDEV $\ominus$

Vasoreactive test



MIST



Diuretics

	◆ Loop diuretics: Furosemide, Bumetanide, Ethacrynic acid, Torsemide
	◆ Thiazide diuretics: Hydrochloro / Chlorothiazide, Chlorthalidone, Indapamide, Metolazone
	◆ Aldosterone blockers: Spironolactone, Eplerenone / Can / Finerenone
	◆ ENaC blockers: Triamterene, Amiloride
	◆ Carbonic anhydrase $\ominus$: Acetazolamide (i.v or oral) / Dorzolamide / Brinzolamide
	◆ Osmotic Diuretics: Mannitol, Urea, Glycerine

◆ Loop diuretic/ High ceiling diuretics

- They act on thick part of ascending limb of loop of henle, symporter system.
- Bumetanide (most potent), Furosemide (max vasodilatory prop), Torsemide (long t_{1/2})
- They have vasodilator properties as they induce PG release

Uses:

- DOC for edema
- Hypertensive emergency, Furosemide is diuretic of choice in Renal failure (Thiazides - ineffective except Metolazone)

Side effects:

- Hypo (.....) emia
- Hyperuricemia, Hyperglycemia, Hyperlipidemia
- Metabolic alkalosis, Ototoxicity (Ethacrynic acid is the most ototoxic - reversible)

◆ Thiazide diuretics: Moderate efficacy diuretics

- Act on proximal part of DCT by $\ominus$ Na⁺Cl⁻ symporter
- Indapamide is primarily used in HTN because of direct vasodilatory effect
- Thiazides are not effective when GFR fall below 30

Uses:

- HTN (DOC), Calcium stones, Rx of nephrogenic DI

Side effects: Same as loops except Hypercalcemia, Erectile dysfunction

Interactions: Loop and thiazide diuretics enhance digitalis toxicity

◆ K⁺ sparing diuretics:

- Act on CD and $\ominus$ Na - K antiport system which is under the influence of aldosterone. Aldo blockers work from basolateral (interstitial) side of tubular cells

Uses:

- Spironolactone is DOC for edema in
Conn syndrome and Resistant HTN. Its also used in CHF and adjunct to diuretics causing hypokalemia

Resmetirom → Rx of Non-alcoholic steatohepatitis (Nash)

- Amiloride is DOC for Rx of Lithium induced DI, Liddle's syndrome, Cystic fibrosis

SE: Hyperkalemia, Metabolic..... Gynecomastia (Spironolactone), Impotence

Gynecomastia drugs: Digoxin, INH, Spironolactone, Cimetidine, Ketoconazole, Oestrogen

◆ Carbonic anhydrase $\ominus$:

- They act on PCT, $\ominus$ the enzyme carbonic anhydrase $\ominus$ $\text{Na}^+ - \text{H}^+$ antiport system causing loss of Na^+ & Bicarb in urine (alkaline urine)

Uses:

- Rx of mountain sickness (DOC), glaucoma, metabolic alkalosis, alkalinization of urine

Side effects:

- They are sulphonamide derivative can cause hypersensitivity reaction (Others: Sulfonylureas, Sulfonamides, Furosemide, Hydrochlorothiazide)
- Hypokalemia (max among diuretics), Hyperammonemia (C/I in Liver failure), Metabolic acidosis

- ◆ Osmotic diuretics: Mannitol is DOC foredema, Acute congestive glaucoma

C/I: Cerebral Hemorrhage, Pulmonary edema, Cardiac failure

Anti-Diuretics

- ◆ Vasopressin agonists

- ⇒ Vasopressin : Non-selective $\oplus$ V1 & $\oplus$ V2 receptors agonist
- ⇒ Terlipressin: selective V1 $\oplus$ - DOC for bleeding esophageal varices
- ⇒ Desmopressin: selective V2 $\oplus$
- DOC for Rx of & in children
- Also used in Rx of VW type I disease and Hemophilia A (quickly develops tachylaxis)

- ◆ Vasopressin antagonists

- ❑ Non-selective $\ominus$ - Conivaptan
- ❑ Selective V1 $\ominus$ - Relcovaptan used for Raynauds disease & tocolysis
- ❑ Selective V2 $\ominus$ - Mozavaptan, Tolvaptan (most potent but hepatotox) is DOC for SIADH

CONGESTIVE HEART FAILURE

- ◆ Acute CHF

- ◆ Inotropes: Dobutamine (DOC), Dopamine
- ◆ Pulmonary edema: furosemide (DOC)

- o **Inodilators:** Inamrinone, Milrinone, Vesnarinone, Enoximone, Levosimendan (Ca⁺ sensitizer)
- MOA: They ⊖ PDE-3 that ↑ C-AMP giving inotropic affect & vasodilatory effect
- o **Natriuretic peptide analogs:** Nesiritide (BNP analog), Carperitide (ANP analog)
- They cause vasodilation along with natriuresis
- ◆ Chronic CHF

- ◆ **ACEI/ARBS:** Candesartan & Valsartan are preferred
β-blockers: Carvedilol, Metoprolol, Bisoprolol
- ◆ **Vasodilators:** Nitrates, Hydralazine, CCB (Non-DHP), Vericiguat
- ◆ **Aldosterone ⊖:** Spironolactone, Eplerenone V2 ⊖ : Tolvaptan
- ◆ **Ivabradine**

- ◆ **Neutral endopeptidase (neprilysin) ⊖:**
- ◆ **Angiotensin receptor Neprilysin inhibitor (ARNI):**
Sacubitril + Valsartan is more effective in decreasing mortality as compared to ACEI/ARBS alone
- ◆ **Vasopeptidase ⊖:** Omapatrilat, Sampatrilat ⊖ both ACE & NEP
SE: Angioedema

Drugs ↓ mortality:

Super: Sacubitril, **Cool:** Canagliflozin (SGLT-2 ⊖)
B ⊖, Hydralazine+IDN, **ACEI/ARB,** Ivabradine
Saab Spironolactone



o Cardiac glycosides (Digoxin, Digitoxin, Oubain)

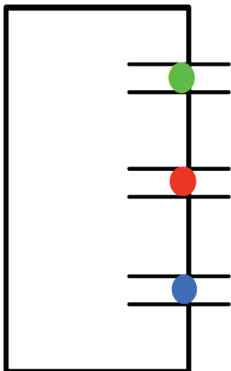
- MOA: vagomimetic action (anti - arrhythmic) → use : At fib
- ⊖ Na-K ATPase (inotrope) → use : CHF

Digitoxin dose adjustment required in liver failure
 Digoxin dose adjustment required in renal failure
 Half-life : 40 hrs, High Vd

- SE: Earliest & MC - Nausea, Vomitting, Gynecomastia, Xanthopsia, Hyperkalemia, Arrhythmia (MC-Ventricular bigeminy, Most characteristic- Non-paroxysmal atrial tachycardia with AV block

- Digoxin toxicity Rx -
 Mild (K⁺), Severe (Digibind dig antibody),
 Ventricular arrhythmia - Lidocaine (DOC)

Cautions: Renal & Thyroid dysfuncton, Myocarditis, MI, WPW synd, ↓ (K,Mg), ↑ Ca,
 P-glycoprotein ⊖ like Verapamil, Amiodarone, Quinidine, Clarithromycin, Itraconazole, Navir (Ritonavir), Erythromycin
 (Mnemonic: Vagcine)



ANTI- ANGINAL DRUGS

Nitrates → $\oplus$ NO → Guanylate cyclase → Smooth muscle relaxation (v.d)

Venodilators → ↓ Cardiac Workload

♦ **Nitrates:** NTG (GTN), Isosorbide mono/dinitrate, Amyl nitrite, pentaerythritol tetranitrate

- ☐ Nitroglycerin (s.l) is DOC for of angina
- ☐ Pentaerythrol (longest acting), Amylnitrate by inhalational route (shortest)
- ☐ All nitrates undergo extensive 1st pass metabolism except IMN (100 % bioavailability)

Uses:

CHF, Angina, MI (CAM)

Biliary colic pain, Achalasia, Cyanide tox (BAC)

SE: Orthostatic hypotension, Headache, Tolerance

Cyanide tox Rx: Amyl nitrite + Na thiosulfate / Hydroxycobalamin

♦ **Nicorandil:** K channel opener in myocardium causing coronary vasodilation, Also releases NO

♦ **Fasudil:** Rho kinase (ROCK) $\ominus$: coronary vasodilation and cerebral vasodilation

♦ **β -blockers:**

- ☐ Used in the long-term prophylaxis of angina and this class of drugs decrease mortality
- C/I in vasospastic/variant angina (DOC -CCB)

♦ **Calcium channel blockers:** Verapamil, Diltiazem

♦ **Ivabradine:** It $\ominus$ funny Na channels SE: Visual disturbances

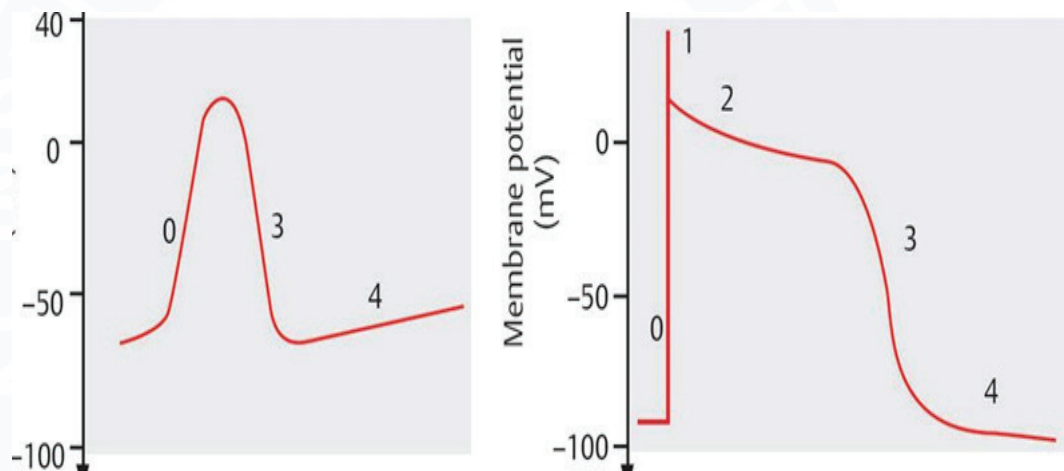
♦ MISCELLANEOUS

- ♦ Ranolazine : ⊖ Late Na Channels
- ♦ Partial fatty acid oxidation (pFox) ⊖: Trimetazidine
- ♦ Dipyridamole: it ⊖ PDE & adenosine agonist. SE:

NSTEMI: Morphine, O₂, Nitrates, Aspirin

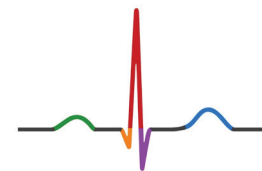
STEMI: Streptokinase, O₂, Nitrates, Aspirin, Morphine

ANTI-ARRHYTHMIC DRUGS



Singh Vaughan - Williams Classification

- Class I (Na channel blockers) ———— { Ia
- Class II (Beta blockers) ———— { Ib
- Class III (Potassium channel blockers) ———— { Ic
- Class IV (CCB - Non-dihydropyridines)
- Class V (Miscellaneous)



◆ Class Ia Drugs:

They are useful in V.tach and fibrillation

As they $\ominus$ K channels risk of Prolonged QT interval (Torsade's)

➤ Procainamide

➤ Quinidine-

It also has anti-malarial effects

Its derived from Cinchona plant (SE: Cinchonism)

➤ Disopyramide - SE: $\ominus$ Muscarinic receptor

◆ Class Ib drugs:

➤ Lidocaine -Digoxin induced Ventricular arrhythmias (DOC), MI

➤ Mexilitine (lidocaine analog), Phenytoin, Tocainide

◆ Class Ic drugs: Propafenone SE: B.C,
Flecainide (DOC- Px of WPW) SE: blurred vision

◆ Class II drugs:

-B -blockers - DOC for rate control in A.fib & flutter,
catecholamine induced arrhythmia (ex - phaeochromocytoma) &
LQTS (long QT synd)

Propranolol (Class 1 & 2), Sotalol (2 & 3)

◆ Class III drugs:

Bretylum (defibrillator), Sotalol, Dofetilide,

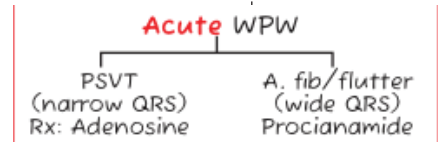
Ibutilide, Vernakalant, Amiodarone (Beautiful Sexy DIVA)

Class III drugs can cause Torsades (Max with Ibutilide, Min with Amiodarone)

➤

➤ Longest acting antiarrhythmic

➤ It also $\ominus$ Na, α , β , calcium receptors so it's wide-spectrum antiarrhythmic (DOC for Ventricular arrhythmias, rhythm control in A. fib & flutter)



Side effects:

Better: blue skin discoloration (Blue Man Synd /)

Check: corneal deposits (whorl like)

LFT: liver toxicity

PFT: pulmonary fibrosis

TFT: thyroid (Hypo>Hyper)

Now: neuropathy

Please: photosensitivity

◆ Class IV drugs:

Non dihydropyridines (Verapamil and Diltiazem)

◆ Class V (Miscellaneous) drugs: ADAM

- **Adenosine:** has $t_{1/2}$ of 5 sec (taken up by cells if given via slow infusion hence rapid infusion in central veins is given).

It's the DOC for Rx of Acute SVT/PSVT

(In Asthma/COPD: Verapamil is DOC for Acute SVT/PSVT)

SE: Hypotension, Bronchoconstriction

Vagal maneuver for PSVT works on M2 receptor

- **Digoxin**
- **Atropine:** DOC for bradycardia and AV block
- **Magnesium sulphate:** DOC for Torsade de pointes and eclampsia in pregnant women

ANTI-HYPERLIPIDEMICS

HMG-CoA reductase ⊖: Lava/prava/simva/atorva/rosuva/pitava **statin**

MOA: ⊖ the rate-limiting enzyme for cholesterol synthesis. Decrease in the plasma cholesterol increases the LDL receptor production hence decreasing the LDL

- ↓↓↓, ↑, ↓
They are DOC for dyslipidemia with raised LDL
- They also have pleiotropic effects (Anti-platelet/coagulant, Vasodilatory and Anti-inflammatory)
- Rosuvastatin is the longest acting, Pitavastatin (most potent), Atorvastatin is safe in renal failure

SE: Hepatotoxic, Myalgia, myopathy (max with simvastatin), C/I in pregnancy and children

Fibric acid derivatives (fibrates): Gemfibrozil, clo / feno / beza.....

fibrate

- (+) PPAR-α and ↑ LPL (Lipoprotein lipase) → ↓ TGL
 - ↓↓↓ TGL, ↓ LDL, ↑ HDL (DOC for hypertriglyceridemia)
 - Side effects: Myopathy and also ↑ statin induced myopathy, Cholesterol gallstones
- ♦ **Niacin (nicotinic acid-Vit B3):**
- It ⊖ lipolysis (↓ HSL - hormone sensitive lipase) → ↓ free fatty acids for triglyceride synthesis in liver
 - ↓↓ LDL, ↑↑, ↓ TGL
 - Side effects: PG mediated(use NSAIDS), Insulin resistance

◆ **Bile acid resins(sequestrants):** Cholestyramine, colestipol, colesevelam

They ⊖ the enterohepatic circulation of bile acids, decreasing the same in liver → ↑ in bile acid synthesis from cholesterol in the liver

- DOC to ↓ LDL in pregnancy & children
- ↓ cholesterol ↑ LDL receptors and ↓ LDL and ↑ HDL
- Side effects: GI upset, ↓ absorption of other drugs and fat-soluble vitamins

◆ **Ezetimibe:** Prevents intestinal cholesterol absorption (⊖ Niemann Pick C1 like 1-NPC1L1 protein) ↓ LDL

◆ **PCSK-9 ⊖ :** Evolocumab, Alirocumab
They increase LDL receptors and ↓ LDL

◆ **PCSK-9 ⊖ & Niacin has shown to reduce lipoprotein (a) / LP (a)**

- Niemann - Pick disease : Arimoclomol (Rx of neurological symp)

CENTRAL NERVOUS SYSTEM PHARMACOLOGY



Active space

MIST

ANTI-EPILEPTIC DRUGS

- ◆ **Sodium valproate:** Wide spectrum anti-epileptic that $\ominus$ Na^+ , T-type of Ca^+ channels, $\uparrow$ GABA, $\downarrow$ Glutamate
- DOC for GTCS (Grand-mal), Myoclonic, Atonic, Absence seizures (Petit-mal), Lennox Gastaut & Dravet syndrome
- Also used in BPD (Rapid cyclers), Acute mania, Px of migraine

Side effects:

V: Vomitting, Nausea (MC),

A: Alopecia/curling

L: Liver tox (C/I < 2 years of age, Rx: Carnitine),

P: Pancreatitis, PCOS,

R: Rash,

O: Obesity,

A: Ataxia (In tox),

T: Teratogenic (NTD), Tremors,

E: Enzyme $\ominus$ (Most CNS drugs are enzyme inducers)

- ◆ **Lamotrigine:** Wide spectrum anti-epileptic that $\ominus$ Na^+ , Ca^+ channels and $\downarrow$ glutamate, $\uparrow$ GABA
- 1st line for Partial seizures, GTCS. It's one of the least teratogenic anti-epileptic. It's also used in absence seizures, BPD
- SE: Syndrome
- ◆ **Ethosuximide:** MOA: $\ominus$ only T type of Ca^+ channels so used in absence seizures
- ◆ **Phenytoin:** MOA: $\ominus$
- Uses: seizures, Digoxin induced VT

- Its an enzyme inducer, which is highly protein bound so can have interactions with other plasma bound drugs like valproate
- It shows both 1st order and zero order kinetics & belongs to class Ib anti-arrythmics
- Side effects: Very HOT MAAL

- ⇒ **Very:** Vit K,D,Folate (B9) ↓ (hemorrhagic disease of new born, Osteomalacia, Meg Anemia)
- ⇒ **Hypertrophy of gum** (.....), Hirsutism, Hyperglycemia (⊖insulin release) Purple glove synd
- ⇒ **Osteomalacia**
- ⇒ **Teratogenic** (Fetal-hydantion syndrome) characterized by cleft lip & palate, hypoplastic phalanges, microcephaly
- ⇒ **Megaloblastic anemia**
- ⇒ **Ataxia, Nystagmus, Diplopia** (In toxicity)
- ⇒ **Arrhythmias** (In toxicity), Acne
- ⇒ **Lymphadenopathy**

◆ **Carbamazepine:** MOA:⊖

- DOC for Trigeminal Neuralgia,
- It's can also be used in BPD, Diabetes insipidus
- It is a potent enzyme inducer

Side effects:

- ⇒ Most of them are d/t + Carbamazepine except hyponatremia (MC with oxcarbamazepine)
- ⇒ Agranulocytosis so C/I with Clozapine, Aplastic anemia, ADH ↑ (SIADH), Autoinduction, Ataxia
- ⇒ Hypersensitivities like SJS

◆ **Levetiracetam:** binds to SV2A (Synaptic Vesicle) 1st line in GTCS, Focal seizures, JME and is DOC in pregnancy

Topiramate: $\ominus$ Na⁺/opens K channels, Glutamate receptors (AMPA, Kainate), $\oplus$ GABA-A receptors

- Uses: Craving ↓, Obesity, Migraine Px, BPD
- SE: Nephrolithiasis (-CA), weight loss
- ◆ **Zonisamide** - $\ominus$ Na⁺ and Ca⁺ channels approved only for partial seizures, SE: Nephrolithiasis
- ◆ **Lacos / Rufinamide** - Na⁺ blocker for partial seizures
- ◆ **Felbamate:** MOA: NMDA receptor $\ominus$ used in partial and generalized seizures SE: BMS
- ◆ **Per / Terampanel** - AMPA blockers approved for partial seizures
- ◆ **Gabapentin & Pregabalin**
 - MOA: ↑ exocytosis of GABA, DOC [Diabetic neuropathy, Post-Herpetic Neuralgia, Restless leg syndrome]
- ◆ **Tigabine:** reuptake $\ominus$ of GAT-1 for GABA used in partial seizures
- ◆ **Vigabatrin:** irreversible $\ominus$ of GABA transaminase (metabolizes GABA) SE: visual field defects
- ◆ **Ezogabine:** K⁺ channel opener used for partial seizures. SE: Blue pigmentation of lips, nails etc

Active space

Status epilepticus	(DOC)
Febrile seizures	Rectal (DOC) Intranasal midazolam
Pregnancy, Eclampsia	Levetiracetam (DOC)/Lamotrigine MgSo4
GTCS, Absence, Myoclonic, Atonic, LGS, Dravet	 Lamotrigine
Focal/Partial seizures Focal (Elderly)	Carbamazepine/Oxcarbamazepine Levetiracetam, Lamotrigine
Infantile spasms/Salaam seizures/West syndrome	With tuberous sclerosis (Vigabatrin), W/O tuberous sclerosis (ACTH)

Sedative-Hypnotics

Barbiturates (GABA mimetic)	Benzodiazepines (GABA facilitatory)
↑ duration of opening of GABA mediated Cl ⁻ channels	↑ frequency of opening of GABA channels
Enzyme inducers	Not enzyme inducer
C/I in Acute intermittent porphyria	No C/I
Anterograde amnesia	Anterograde amnesia
Dont have any specific antidote	Antidote : Flumazenil

◆ Benzodiazepines

Long acting (C/I in liver failure) - Dia / Clona / Flura / Flunitra / Nitrazepam

Short acting (Safe in liver failure) - Estazolam, Oxa / Teme / Lorazepam

Uses:

- ❑ Antiepileptic - Lorazepam, Diazepam
- ❑ Hypnosis - ↓ all phases in NREM except II
- ❑ Anti anxiety (Sedatives) - Oral Diazepam, Oxazepam, Lorazepam, Alprazolam
- ❑ Muscle relaxation
- ❑ Pre-anesthetic medications - Diazepam, Midazolam, Lorazepam
- ❑ Alcohol withdrawal - Chlordiazepoxide (DOC) except seizures- Lorazepam
- ❑ Date rape -

◆ Barbiturates

- Long acting - Phenobarbitone
- Short acting - Pentobarbitone, Secobarbitone, Butobarbitone
- Ultra-short acting - Thiopentone, Methohexitone

Active space

Uses :

- Phenobarbitone -used for Rx of seizures in neonates, also used in status epilepticus & Crigler-Najjar type II
- Thiopentone -redistribution phenomenon, used as i.v anesthetic (for induction)
- Methohexitone - used in Electroconvulsive therapy

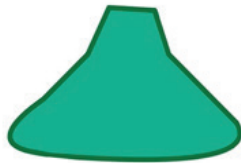
INSOMNIA TREATMENT:

- ◆ Z drugs
 - MOA - selective $\alpha 1$ agonist at GABA-A receptor used for hypnosis & sedation
 - Zaleplon - short acting for sleep induction in insomnia, jet lag
 - Zolpidem - intermediate acting for short-term Rx of insomnia
 - Eszopiclone - long acting for sleep maintenance & long-term Rx of insomnia
- ◆ Antidote for Benzodiazepines & Z drugs is Flumazenil
- ◆ Melatonin agonists (MT1 - sleep, MT2 - circadian rhythm)
 - Ramelteon - Used for sleep induction in insomnia & jet lag
 - Tasimelteon - Used in blind & smith magenis syndrome
- ◆ Dual Orexin (promotor for wakefulness) receptor antagonist: (DORA): Suv / Darid orexant: Used for sleep maintenance

ANTI-ANXIETY

- Benzodiazepines (Ex: Alprazolam) are the mc anti-anxiety drugs & SSRI's are also used
- 5-HT_{1a} agonist: Bus / Ipsa / Gepirone (non-sedative drugs)
- β -blockers -Propanolol

ANTI-DEPRESSANTS



- ◆ ($\ominus$ uptake of NA & 5HT)
- Clomipramine - max 5HT $\uparrow$
- Ami / Nortryptalline - Rx of peripheral neuropathy & also used in smoking dependence and migraine prophylaxis
- Amoxapine - has additional D2 blocking effect (antipsychotic derived from amoxapine k/a Loxapine)
- Doxepin - strong antihistaminic activity
- Imipramine - nocturnal enuresis

Side effects:

- ❑ TCAs additionally block α_1 , H1 and M (HAM)
- ❑ M block produces dry mouth, blurring of vision, urinary retention & constipation, & sedation (H1 & M1 block), Postural hypotension d/t α block, Weight gain d/t 5HT block
- ❑ TCA toxicity produces cardiac arrhythmias (Rx-Lidocaine) & convulsions (Rx-Diazepam) and acidosis (Rx- Sodium Bicarbonate)
- ◆ **SNRI (Serotonin NE reuptake $\ominus$)** - Used for severe depression
 - Venlafexine
 - Duloxetine and Milnacipran for Fibromyalgia.
Duloxetine is also used for stress incontinence or neuropathic pain
- ◆ **Selective Serotonin Reuptake $\ominus$ (SSRI)**
 - DOC for mild to moderate depression, OCD, phobias, PTSD, eating disorders like bulimia nervosa & anorexia nervosa, premenstrual tension syndrome (PTS), panic attacks, GAD
 - Fluoxetine is the longest SSRI so preferred in most cases
 - Fluvoxamine shortest SSRI. Others are Paroxetine, Citalopram, Escitalopram, Sertraline

Side effect:

- $\oplus$ 5HT₂: Anxiety initially followed by anxiolytic effect after 2 weeks, Delayed ejaculation so SSRI can be used in Rx of premature ejaculation
- $\oplus$ 5HT₃: (MC side effect),
- $\oplus$ 5HT₄ in GIT: loose stools

◆ MAO- $\ominus$

- Non-selective (MAO -A & MAO -B $\ominus$) - Tranylcypromine, Phenelzine, Isocarboxazid
- Selective MAO-A $\ominus$ - Moclobemide
- Selective MAO-B $\ominus$ -Sele / Rasa **geline**

Side effects:

- ❑ MAO $\ominus$ should be avoided with other drugs that $\uparrow$ 5HT (SSRI, SNRI, TCA, Opioids, Amphetamines etc) d/t risk of Serotonin syndrome
- ❑ MAOI inhibit metabolism of Meperidine leading to seizures, Cheese reaction, Hepatotoxic

◆ Atypical antidepressants
5HT₂ $\ominus$: Trazodone SE: Priapism

NSSA (NA specific Serotonergic Antidepressant): Mirtazepine
 $\ominus$ α_2 , 5HT(2,3). Advantage: Less sexual side effects

NDRI (NA Dop Reuptake Inh): Bupropion - used in smoking cessation

New drugs:

Esketamine - nasal spray,

Brexanolone(Allopregnanolone), Zuranolone (oral)-Rx of Post-partum depression

BIPOLAR DISORDER

BDNF (Brain derived neurotrophic factor)

- ◆ **Mania** - Lithium is DOC for Px, Anti-psychotics (DOC in preg), Anti-epileptics like Valproate (DOC for rapid cyclers in BPD), Topiramate, Carbamazepine are also used
- ◆ **Depression** - Lithium is DOC for Px, Lamotrigine is also indicated
- ◆ **Lithium**
 - **MOA:** $\ominus$ IMP which $\downarrow$ IP3 required for BDNF synthesis, also $\ominus$ GSK-3 pathway ($\ominus$ metabolism)
 - It acts as mood stabilizer (controls both depression and mania)
 - Thiazide & K sparing diuretics, (enhance tubular reabsorption of Lithium) ACEI, NSAIDS $\uparrow$ the risk of lithium toxicity

Side effects:

Leukocyte increase,
Tremors,
Hypothyroidism,
Inc urine (Li induced Nephrogenic DI),
Mouth (burning), Mother c/I (.....),
Lithium is also C/I in Renal failure

- It is a drug with low T.I so regular TDM is required

Therapeutic range: 0.5- 1.2 mEq/L

Toxicity: >1.5 mEq/L

Dialysis: >4 mEq/L

ANTI-PSYCHOTICS (NEUROLEPTICS)

◆ Typical antipsychotics: $D_2 \ominus$

Low potency	High potency		
Phenothiazines	Butyrophenone	Thioxanthene	Phenothiazines
Chlorpromazine	 (Max EPS)	Thiothixene	Fluphenazine
Thioridazine (Minimum EPS)	Droperidol	Penthixol	Trifluoperazine

◆ Atypical anti-psychotics ($5HT-2A \ominus > D_2 \ominus$):

Ari / Brexipiprazole, Clo / Olanza / Quetiapine,
Paliperi / Risperi / Ziprasidone

USES of Antipsychotics: Anti-emetic, Psychosis, Chorea, Ticks, Hiccups

Clozapine is DOC for Resistant Schizophrenia and reduces suicidal tendencies.

SE: Agranulocytosis (dose independent), Convulsions (high dose),
 $\ominus$ M (sedation), Wet-pillow syndrome

Aripiprazole is also partial agonist at D_2 receptors, $5HT_{1A}$ receptor

Pimavanserin $\ominus$ $5HT_2$ only, NO $D_2 \ominus$ (No EPS / No prolactin $\uparrow$)
for Parkinson Rx induced Psychosis

- Typicals (Haloperidol, Chlorpromazine) Atypicals (Quetiapine Ziprasidone) Cause QT prolongation
- Thioridazine cause retinal degeneration, Chlorpromazine cause cholestatic jaundice

- is the most potent atypical with strong D2 block (EPS) and MC to cause hyperprolactinemia,
- Quetiapine, Clozapine causes cataract

EPS (extrapyramidal side-effects) are seen mostly with high potency
Typicals, Atypicals

Acute muscular dystonia (<1 week): presenting with Torticollis and oculogyric crisis (Rx -central anticholinergic-DOC, Antihistaminics like Promethazine, Diphenhydramine)

Drug induced parkinsonism (5d-1month): Rx - central anticholinergic

Akathasia (MC EPS - 5d-2month): Rx- DOC-Propanolol, BZD are add ons

Tardive dyskinesia: Rx- change to Atypicals, VMAT2 ⊖ like Valbenazine, Deutetrabenazine

Neuroleptic malignant syndrome - presents with life threatening hyperthermia. Rx -i.v Dantrolene (DOC), Bromocriptine (D2 agonist)

Dantrolene (Ryanodine rec ⊖) is also DOC in Rx of Malignant hyperthermia (due to excess Ryanodine receptor action releasing excess Ca⁺) that is caused by Halothane or Succinylcholine

- Other receptor block (HAM): Thioridazine>Haloperidol
H1 & M block causes sedation & anticholinergic side effects,
H1 & 5HT block leads to weight gain max with olanzapine,
clozapine least with Aripiprazole and Ziprasidone
α1 block causes postural hypotension

ANTI-PARKINSONS DRUGS

Dopamine precursor:

- MOA: It's a prodrug of dopamine that is metabolized by Dopa decarboxylase (DD) into Dopamine (Best drug for parkinson's)
- Carbidopa & Benserzide are lipid insoluble, peripheral DD ⊖
- Levodopa is always combined with peripheral DD ⊖ but Pyridoxine (B6) is C/I
- SE: On-Off phenomenon (Rx: MAO-B⊖, COMT⊖, D₂⊕), Psychosis (Rx: Pimavanserin), Vomitting (Rx: Domperidone)
- ◆ COMT⊖
- Tolcapone ⊖ both in periphery and centre but its hepatotoxic, Enta / Opicapone ⊖ only in periphery
- ◆ MAO-B⊖
- Selegiline / Rasagiline, Safinamide (For On-Off effect)
- They prevent neurodegeneration
- Transdermal selegiline is used for Rx of depression

◆ D2 agonists

- **Ergot derivatives :**
Side effects : Peripheral vasoconstriction that leads to gangrene, fibrosis, Valvular Heart defects
- **Non-Ergot derivatives :** Pramipexole, Ropinirole, Rotigotine are longer acting than Levodopa hence preferred in On-Off phenomenon Parkinsons (DOC), Restless leg syndrome

- ◆ Apomorphine is an opioid with D2 agonist activity

Drug induced parkinsonism:

- ◆ **Central anti-cholinergic's:** Benztropine, Benzhexol (trihexyphenidyl), Biperiden are DOC
- ◆ **1st generation antihistamines:** diphenhydramine and promethazine
- ◆ **Amantadine** increases dopamine by increasing release. It also has anti- cholinergic effect. SE- Livedo reticularis, Ankle edema



◆ Rx of Alzheimer's disease:

- **Cholinergic drugs :** Donepezil (DOC), Rivastigmine, Galantamine, Tacrine (no longer used as it is hepatotoxic)
- **β amyloid $\ominus$:** Adacanumab, Lecanemab, Donanemab
- **NMDA blocker :** Memantine
- **Nootropics (Memory enhancer) :** Piracetam

◇ Rx of Huntington's chorea:

- VMAT $\ominus$: Tetrabenazine (DOC).
- D₂ $\ominus$: Atypical (aripiprazole), Typical (Haloperidol)

◇ Rx of Amyotrophic lateral sclerosis:

- Riluzole (NMDA blocker), Baclofen -DOC for spasticity, Edaravone (free radical scavenger) Tofersen ($\ominus$ SOD1 protein)

◇ Rx of Retts Syndrome: (Cerebroatrophic Hyperammonemia)
Trofinetide

◇ Smoking dependence:

- Varenicline, Cytisinicline (cytisine), Nicotine inhalers & Bupropion are 1st line drugs
- Clonidine & Nortriptyline are 2nd line drugs

OPIOIDS



- The opioid receptors are of 3 subtypes i.e μ , κ , δ acted upon by endogenous opioids called as endorphins, dynorphins, enkephalins respectively
- μ : Miosis, Urinary retention, Sedation, Constipation, Analgesia, Respiratory depression, Increased muscle rigidity, Euphoria (Mnemonic: MUSCARI μ E)
- κ : Dysphoria, Analgesia, Resp depression, Diaphoresis, Sedation (Mnemonic - DARDS)
- δ : Analgesia
- μ -agonists produce dependence d/t euphoria. Tolerance develops to all except Miosis, Constipation, Convulsions. They also produce sedation & nausea vomiting by $\oplus$ CTZ

Active space

◆ Classification:

- **Full agonists** - Morphine, Meperidine (Pethidine), Methadone, Codeine, Tramadol, Loperamide
- **Mixed ($\oplus\kappa$, $\ominus\mu$)** - Dezocine, Nalbuphene, Butarphanol, Buprenorphine (partial agonist at μ and $-\kappa$), Pentazocine are ($+\kappa$, $-\mu$) (Mnemonic: Do Not mix Beer & Paani)
- **Pure antagonist** - Naloxone, Naltrexone, Nalmefene, Alvimopan & Methylnaltrexone are peripheral

◆ It's used by both oral and parenteral routes as an analgesic in labour, cancer patients, MI. Its also useful in pulmonary edema as it also produces v.d (d/t Histamine release)

➤ **S/E:** NV ($\oplus$ CTZ), Hypotension, B.C and Pruritis (d/t Histamine), Resp depression, Constipation

➤ Its C/I in head injuries ($\uparrow$ ICT), bronchospastic disorders, Renal failure, Biliary colic pain

◆ **Meperidine (pethidine)** Its DOC for post-anesthetic chills and infusion related chills

MAOI $\ominus$ Meperidine metabolism & $\uparrow$ neurotoxic (seizure) metabolite normeperidine so they are C/I, also Meperidine and MAOI both $\ominus$ Serotonin metabolism leading to Serotonin syndrome

➤ **and Diphenoxylate** are DOC for diarrhea associated with IBS. New opioids like Fedotozine (κ agonist) & Eluxadolone (μ & κ agonist) can also be used

- ❑ Diphenoxylate (Cross BBB) causing Euphoria, so combined with Atropine to ↓ dependence
- **Methadone/Buprenorphine** : used for de addiction as replacement followed by Naltrexone to prevent relapse
- **Codein/Non-Opioids (Dextromethorphan,Noscapine)**: used as an antitussives
- **Tramadol/ Tapentadol**: analgesic for moderate pain, it also ⊖ reuptake of NA & 5HT
- **Oxy / Hydrocodone** used as analgesics
Suzetrigine (Non-opioid) : used as analgesic
- **Fentanyl group**: Alfentanyl, Fentanyl (100x morphine), Sufentanyl (1000x) are used in anesthesia
Also available as analgesic patch
SE: Truncal rigidity (Wooden Chest syndrome)
- **Papaverine**: used for Erectile dysfunction
- ❑ **Pure Antagonist**

Opioid Toxicity triad: Pin-point pupil, respiratory depression, unconsciousness

- (Short acting) is a DOC for opioid toxicity
Naltrexone (Long acting) for maintenance & to prevent relapse after de-addiction
- Naltrexone + Buprapion for obesity

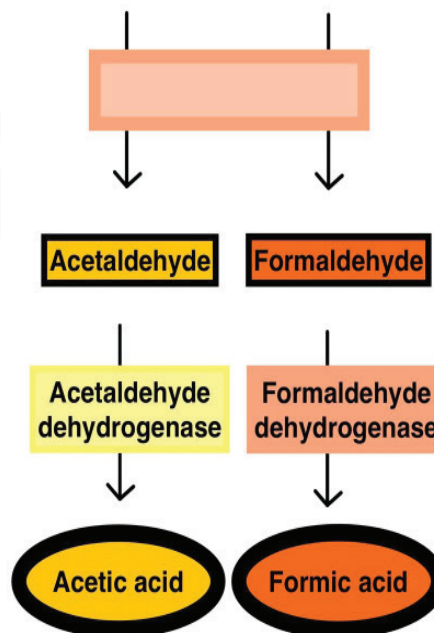
Active space

- Alvimopan, Methylnaltrexone, Naldemedine, Naloxegol: Rx of Opioid induced constipation
- ◆ **Alcohol**
 - **Toxicity** - DOC is ($\ominus$ alcohol dehydrogenase), Ethanol for Methanol poisoning
 - **Withdrawal** - Benzodiazepines are DOC (Diazepam, Lorazepam, Chlordiazepoxide)
 - **Dependence** - Disulfiram (acetaldehyde dehydrogenase $\ominus$), Topiramate, Ondansetron, Naltrexone (Opioid blocker), Acamprostate (- NMDA receptors) decrease cravings in alcoholics

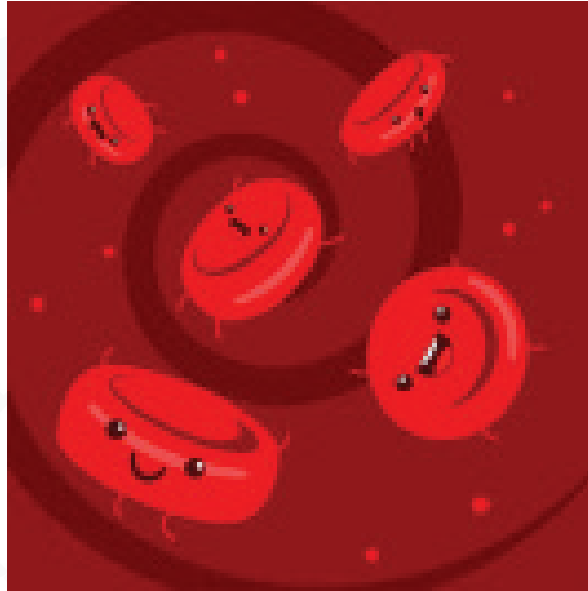
Disulfiram like Rxn: Metronidazole, Cephalosporins (Cefoperazone, Cefotetan, Cefomandole), Chlorpropamide, Griseofulvin

DER (Disulfiram-Ethanol Rxn):

Palpitation, Nausea,
Vertigo, Flushing
Diaphoresis Hypotension



Active space



Blood forming drugs

◆ Iron

- Iron is absorbed in ferrous form from Duodenum, Ferrous Suphate is the oral agent of choice

Substances which ↑ iron absorption are Ascorbic acid (Vit C), Gastric acid (Hcl), Amino acids

Substances which ↓ iron absorption are Tetracycline (Chelating agent), Alkalis, Tea

In acute iron poisoning Rx is done by iron chelating agents like Desferroxamine (i.v,i.m)

Chronic iron overload and thalassemia → Desferroxamine, Oral Deferiprone/Deferasirox

SE: i.m causes staining of skin (Z technique is used for prevention)

- BAL (British Anti-Lewisite) is useful in Bismuth, Arsenic, Lead (EDTA also used) but not useful in Cadmium and Fe poisoning

◆ Folic acid: Site of absorption - Jejunum

- Is used in the Rx of megaloblastic anemia and prophylaxis of NTD
- Leukovorin [folinic acid] - is used in Rx of methotrexate poisoning

◆ Vit B12 / Cyanocobalamin: Site of absorption - Ileum

- Used in the treatment of pernicious anemia, gastrectomy etc
- B12 given - i.m or s.c. Orally - for other causes deficiency of intrinsic factor

Growth Factors required for blood formation

1) **RBC:** Epo is the growth factor. Drugs: EPOIETIN α , / β and Darbopoietin, Peginesatide

Uses: Anemia in patients with CRF, Chemotherapy, Zidovudine

Peginesatide - epo receptor agonist for Rx of anemia associated with CRF

Daprodustat → Rx of Anemia d/t CKD in Dialysis

Side effects of EP therapy - Polycythemia & HTN

2) **Platelets:** Thrombopoietin & IL 11 are Growth factor for platelets.

Drugs: Oprelvekin, Romiplostim, El / Ava / Luso **trombopag**

➤ Uses: thrombocytopenia d/t anticancer drugs, ITP

➤ Oprelvekin - is recombinant IL -11

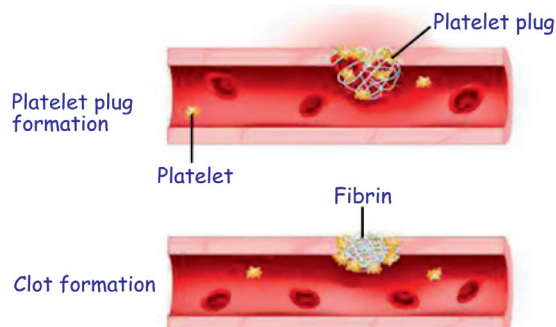
3) **WBC:** G-CSF & GM-CSF are growth factors.

Recombinant G-CSF = Leno / Filgrastim,

Recombinant GMCSF = Mol / Sargramostim

➤ Uses: These agents are used in Rx of leucopenia induced by cancer chemotherapy

ANTI - PLATELET DRUGS

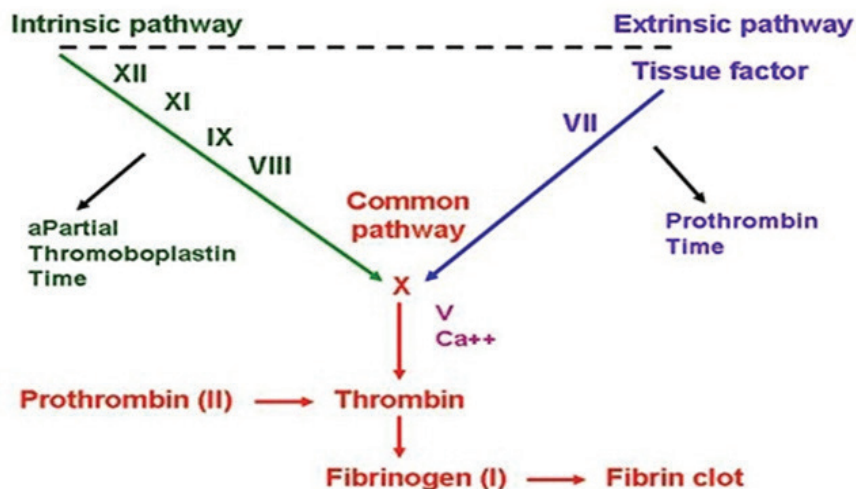


- ◆ Anti - PLT drugs are used in prophylaxis of TIA, Stroke, MI, Angina, Per Vascular diseases
- ◆ TXA₂ $\ominus \rightarrow$ (40-325mg)
- ◆ ADP (P₂Y₁₂) $\ominus \rightarrow$
 Irreversible (Clopidogrel / Prasugrel, Ticlopidine),
 Reversible (Ticlopidine / Cangrelor).
 Omeprazole (.....) is C/I with Clopidogrel

 Prasugrel is fastest acting but high risk of bleeding so C/I in stroke
- ◆ GP IIb/IIIa $\ominus \rightarrow$ Abciximab(safe in renal failure), Eptifibatide, Tirofiban (Strongest anti PLT drug as they block aggregation induced by all agonists)
- ◆ PAR-1 (Protease activated receptor) $\ominus \rightarrow$ Vorapar / Atopaxar
 (PAR-1 inhibition $\ominus$ thrombin receptors on platelets giving anti-platelet effect)
- ◆ Phosphodiesterase $\ominus \rightarrow$ Dipyridamole, Cilostazol (used in PVD, Intermittent claudication)
 - ❑ Cilostazol & dipyridamole are PDE 3 $\ominus$ & $\uparrow$ CAMP (anti-aggregant effect)
 - ❑ Dipyridamole causes / Coronary steal phenomenon (not used in MI)
- ★ SE: Bleeding is the main side effect with all anti - PLT drugs

ANTI - COAGULANTS

USES: DVT, Pulm embolism, Cancer induced thromboembolism, PCI, A.Fib, Prosthetic Valves etc



◆ Oral:

❑ Indirect: Vit K ⊖: Warfarin, Dicumarol (coumarin)

❑ Direct: Direct oral anti-coagulants (DOAC)
They are DOC for Px of DVT (Rx: LMWH)

❑ Factor Xa ⊖ - Rivaroxaban / Apixiban / Edoxaban / Betrixaban
Antidote for Xa ⊖: Andexanet alfa

❑ Direct thrombin ⊖ - Dabigatran (Antidote-Idarucizumab)

● Ciraparantag: universal antidote for all anti-coagulants except Warfarin

◆ Parenteral:

❑ Direct thrombin ⊖ - Lepirudin / Hirudin / Bivalirudin, Argatroban

❑ Indirect thrombin ⊖ - Heparin, LMWH (Enoxaparin / Dalteparin / Tinzaparin), Fondaparinux

- ❑ UFH (Unfractionated) → $\oplus$ AT → $\ominus$ Xa = $\ominus$ IIa
- ❑ LMWH → $\oplus$ AT → $\ominus$ Xa $>$ $\ominus$ IIa
- ❑ Fondaparinux → $\oplus$ AT → $\ominus$ Xa

LMWH are DOC for Rx of thromboses due to DVT, MI, Angina, Pulmonary embolism etc

Heparin (i.v or subcutaneous)	Warfarin (Oral)
$\oplus$ AT → $\ominus$ Xa, $\ominus$ IIa	$\ominus$ vit K dependent factors (II, VII, IX, X, Prot C, S)
Rapid onset of action	Delayed onset (4-5 days)
Monitored by aPTT (~30 sec)	PT (~15 sec) or INR (Target: 2~3) 3-10 : Stop warfarin and restart at low dose >10: Stop warfarin, give vit K
Does not cross placenta (DOC in pregnancy) 1st : UFH / LMWH 2nd : LMWH (best), UFH 3rd : LMWH - switch to UFH near delivery	C/I: Fetal Warfarin syndrome - growth retardation, hypoplastic nose, and hand bones 13-36 weeks
Side effects: Bleeding, Heparin Induced Thrombocytopenia/HIT (DOC - Argatroban), Osteoporosis, Alopecia, Hyperkalemia	Bleeding, dermal vascular necrosis (Purple toe synd)
Antidote -	Inj Vit K + 4-factor complex > FFP if NA then Whole blood

- Warfarin shows kinetics which change from 1st order to zero order
- Warfarin has many interactions with enzyme inducers and enzyme $\ominus$

- Chronic alcoholism may enhance its toxicity d/t deficient production of clotting factors
- Sulfonamides can also precipitate Warfarin toxicity

FIBRINOLYTIC/THROMBOLYTIC AGENTS

USES: thrombolysis in acute MI, severe pulmonary embolism, DVT, Stroke

- 1) **Streptokinase** : It is antigenic, causes allergic reactions, produces neutralizing Abs
- 2) **Recombinant tissue plasminogen activators (tPA)** :
Alte / Rete / Dute / Tenecte
They are non - antigenic and preferred fibrinolytics

SE: Bleeding

Antidotes: Anti-fibrinolytics like Epsilon Amino Caproic Acid (EACA) & Tranexemic acid

They are used for Rx bleeding from Fibrinolytics, Bleeding issues like Menorrhagia, PPH, Hemophilia etc

Sickle cell Disease Rx :

Hydroxyurea : ↑ HbF, ↓ adhesion, ↓ neutrophils → less sickling

Voxelotor : Stabilizes oxygenated Hb → prevents HbS polymerization

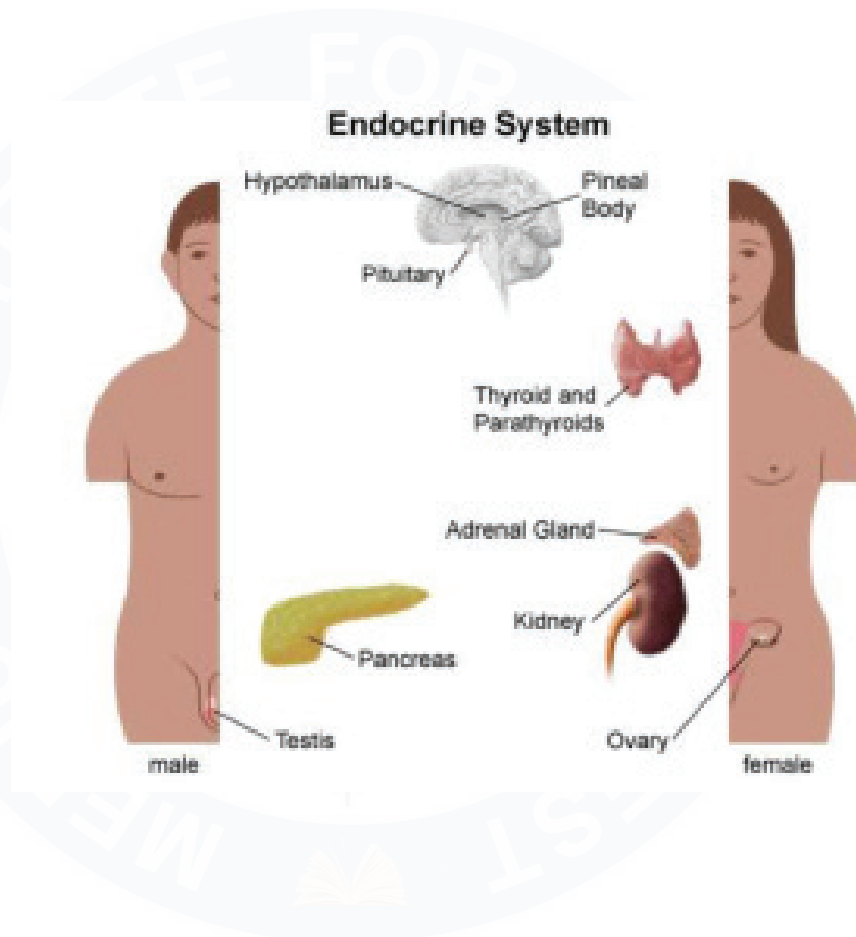
Crizanlizumab : Anti -P- selectin → ↓ cell adhesion & vaso-occlusion

L-Glutamine : Reduces oxidative stress → less hemolysis

Paroxysmal nocturnal hemoglobinuria

Rx : Eculizumab (⊖ C 5)

ENDOCRINE PHARMACOLOGY



DIABETES MELLITUS

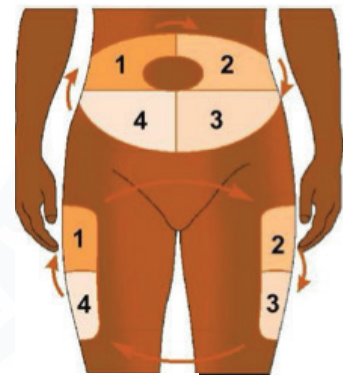
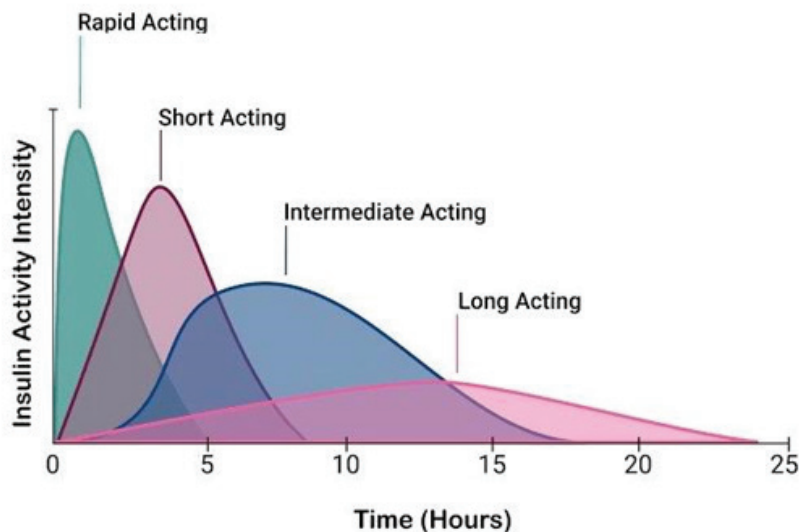
◆ Insulin

- It is an anabolic hormone whose most rapid absorption occurs from abdomen. It acts through receptors

- All insulins are given by s.c route except Regular insulin (i.v)
- Afrezza (inhalational-fastest but ultra-short acting). Its C/I in Smokers, Asthmatics and Lung cancer
- It's the DOC for type I DM, gestational DM, DKA
- Its also used in DM II in later stages and also in hyperkalemia
- MC side effect is hypoglycaemia & lipodystrophy (Atrophy / hypertrophy) at the site of injection



Types:



Rapid : Lispro, Aspart, glulisine

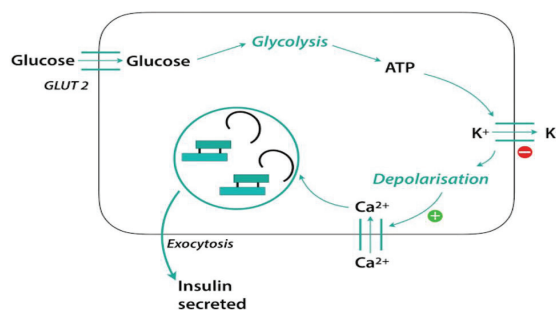
Short: Regular human insulin (Rx : DKA)

Intermediate: Lente, Neutral Protamine Hagedron (NPH)

Long acting (peakless insulins): Glargine, Detemir, Degludec

- Shortest acting: Afrezza (2-3 hrs) Longest acting: Degludec (42hrs)
- Most insulin preparations are available at Neutral pH except Glargine (acidic pH) so Glargine is cannot be mixed with other insulins.
- Zinc is added to insulin to stabilize the hexamer form of insulin
- Lente insulin is combo of Semi-lente (short acting, 30% amorphous powder) + Ultra-lente (long acting , 70% crystalline)

ANTI - DIABETIC AGENTS



Insulin Secretagogues

- ◆ **Sulfonylureas:** MOA: increase insulin release by inhibiting ATP sensitive K channels

- **1st gen:** Chlorpropamide, Tolbutamide, Acetohexamide
- Chlorpropamide is longest acting and Tolbutamide is least potent & shortest acting SFU
- Tolbutamide is hepatotoxic & chlorpropamide can cause SIADH & disulfiram like reaction
- **2nd gen:** Glyburide, Glipizide, Glicazide, Glimipride are more preferred

SE: Weight gain & hypoglycemia

◆ Meglitinides: MOA: same as SFU but less potent

Ex: Rapa / Nateglinide

They are used for PP-hyperglycemia

SE: Weight gain & hypoglycemia (less risk)

➤ GLP-1 agonist: Stimulate insulin release, delay gastric emptying inducing satiety and are given s.c

Ex : Exenatide, Lira / Albiglutide / Semaglutide / Dulaglutide

➤ SE: Nausea, vomiting, wt loss, pancreatitis

✓ GLP-2 agonist: Teduglutide

✓ GLP-1 & GIP (gastric inhibitory polypeptide) analogue: Tirzepatide

Rx of Obesity

- ✓ Lipolysis stimulator: Mirabegron
- ✓ Decreasing appetite: Liraglutide, Phentermine
- ✓ Lipase inhibitor (inhibits absorption): Orlistat
- ✓ Unknown mechanism: Topiramate, Naltrexone, Bupropion

◆ DPP (Dipeptidyl peptidase) 4 ⓪:

➤ MOA: ↑ GLP-1 activity by inhibiting DPP-4

➤ Ex: Sitagliptin / Saxagliptin / Alogliptin / Vildagliptin / Teneligliptin . Linaagliptin

➤ Primarily excreted by Kidney so dose should be decreased in renal failure

➤ SE: Pancreatitis, angioedema, Nasopharyngitis

- ◆ **Biguanide:** Stimulate AMPK leading to inhibition of gluconeogenesis, ↓ insulin resistance, ↓ LDL
- Metformin is DOC for type II DM Rx & Px and can ↓ micro & macrovascular complications
- Its also used in..... reducing anti-psychotic associated weight gain & gestational DM
- SE: lactic acidosis, At high dose ↓ B12 absorption (Megaloblastic anemia), Hypersensitivity reactions in the form of anaphylaxis, angioedema, SJS is seen
- C/I in renal failure (GFR<30) as risk of lactic acidosis increases

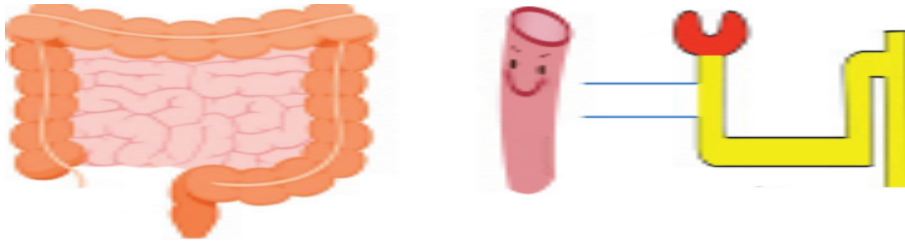
◆ Thiazolidinediones

- MOA: stimulate PPAR, a nuclear receptor that ↑ GLUT 4
Ex: Pio / Rosi**glitazone**
- SE: Osteoporosis (↑ risk of fractures), Macular edema, Pioglitazone increases risk of Bladder cancer

They stimulate ENaC channels causing edema, weight gain
They are metabolized in liver hence they are safe in renal failure

- Other drugs safe in RF: GLP-1 agonist (Dula/Albiglutide), DPP-4 ⊖ (Lina/Teneligliptin)
- Cardiovascular benefits: GLP-1⊕, SGLT-2 ⊖

- ✓ PPAR α+γ agonist: Saroglitazar



- ◆ **Alpha-Glucosidase** ⊖:, Voglibose, Miglitol
 - MOA: This enzyme metabolises starch & disaccharides into glucose
 - SE : Flatulence and Diarrhea. C/I in IBS

- ◆ **SGLT-2** ⊖ : These receptors present in PCT absorb glucose & Na
 Ex: Cana / Dapa / Remo / Empa / Sota **gliflozin**
 Approved for Chronic CHF
 SE: UTI (Urosepsis) & Vaginal infections, Fournier's gangrene, ↑ risk of fractures (canagliflozin), Wt loss

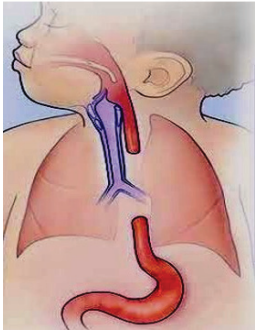
- ◆ **Amylin analogue:**
 - MOA: ↓ glucagon release, delays gastric emptying and induces satiety given along with insulin s.c in both DM type I & II
 - SE: Nausea, Vomitting, Wt loss
 - Bromocriptine - unknown mechanism, Colesevelam - bile acid sequestrant

Weight gain	Sulfonylurea, Insulin, Thiazolidinediones
Weight loss	GLP-1 ⊕, SGLT-2 ⊖, Pramlintide, Metformin
Weight neutral	DPP-4 ⊖, Alpha glucosidase ⊖, Metformin

Diab retinopathy: Anti-VEGF (Bevacizumab, Ranibizumab)

Diab gastroparesis: Metoclopramide, Erythromycin, Domperidone

ANTI - THYROID DRUGS



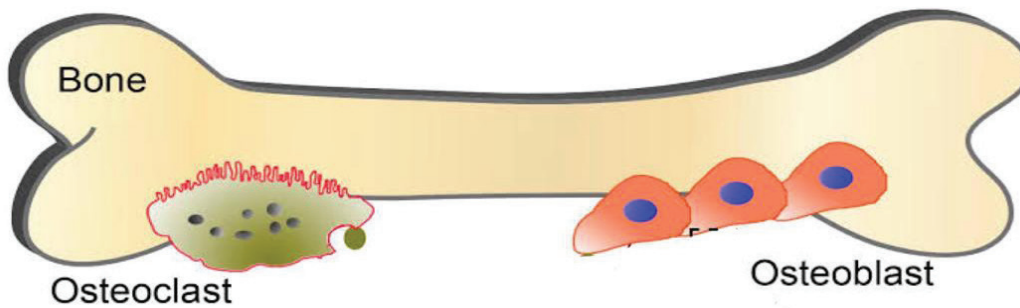
- **Thyroid peroxidase (TPO)** ⊖
- **Carbi (prodrug) / Methimazole** - DOC for hyperthyroidism (Ex: Grave's disease) and also in last two trimesters of pregnancy
SE : Rash (mc), Agranulocytosis
Teratogenic : Esophageal/choanal atresia, cutis aplasia
- **Propylthiouracil**: It ⊖ TPO & also ⊖ peripheral T4 to T3 conversion by inhibiting deiodinase
DOC for 1st trimester & lactation and thyroid storm (TOC - blockers+ PTU). SE: hepatotoxic
- ◆ **Iodides**: KI & Lugol's iodine (5% iodine + 10% KI)

They are used prior to Sx to ↓ size of the gland (make it firm) and reduce vascularity (↓ bleeding)
- ◆ **Radioactive iodine**
 - I 123 emits only γ rays hence used for thyroid scan only
 - I 131 emits β & γ rays that destroys the follicular cells. So its used for Rx of hyperthyroidism in older patients, heart disease, its C/I in pregnancy
Side effect: permanent hypothyroidism, secondary malignancies
- ◆ **Peripheral conversion**⊖: Propranolol (B-Blockers), Prednisone (Steroids) and PTU

THYROID HORMONE REPLACEMENT

- ◆ **Levothyroxine (T4)**: longer acting Rx of hypothyroidism
- ◆ **Liothyronine (T3)**: short acting Rx of Myxedema coma

BONE DRUGS



- **Bisphosphonates:** MOA: ↑ apoptosis of osteoclast ($\ominus$ Farnesyl pyrophosphate synthase in osteoclast)
- ✦ Ex: Alen / Rise / Zolendronate
- Zolendronate (most potent) DOC for Rx of osteoporosis, Rx of hypercalcemia associated with malignancies, Paget's disease
- Drugs like Steroids, Heparin, Thyroxine are associated with osteoporosis
- Given max for 5 years then 1-2 years of drug holiday

Side Effects:

Esophagitis (Prevention: Sit upright for 30 mins after taking medicine, take with full glass of water) Osteonecrosis of jaw, Hypo/hypercalcemia, increased risk of bone fractures

- ◆ **Estrogen & SERM:** MOA: Raloxifene & Estrogen inhibit bone resorption & are indicated for Rx & Px of postmenopausal osteoporosis
- ◆ **Salcatonin:** MOA: Calcitonin analog given Intranasal for Px of Osteoporosis

- ◆ **Remosozumab:** Anti-sclerostin antibody
- ◆ **Denosumab:** MOA: It's a RANK-L antibody inhibiting activation of osteoclasts

Parathormone related drugs

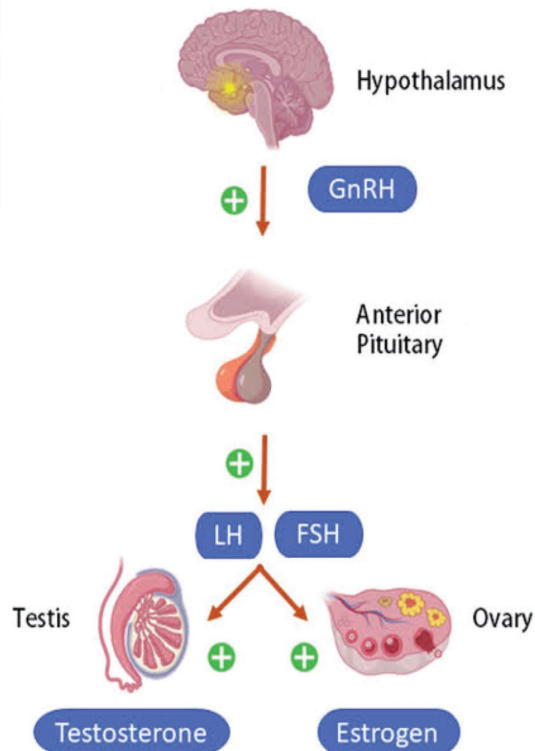
.....: MOA: incomplete PTH (1-34) analog that activates osteoblasts for bone formation

Teriparatide increase osteosarcoma risk

Abolaparotide: PTH related peptide (PTHrP) analogue

Strontium Ranealate: Its only class of drug that acts both on bone formation & resorption

DRUGS OF REPRODUCTIVE SYSTEM



◆ GnRH agonists: Gosa / Busa / Nafarelin, Leuprolide

- At continuous doses there could be worsening of symptoms d/t initial increase in LH/FSH
- Side effects: hot flashes and vaginal atrophy, impotence in males

◆ GnRH antagonists: Gani / Aba / Cetrorelix

They are given s.c only at continuous doses but there is no initial increase in LH/FSH

◆ Estrogen

◆ **Natural:** Estrone (Menopausal), Estrodiol (Reproductive) Estriol (Pregnancy)

◆ **Synthetic:** Mestranol, Ethinylestradiol, Stilbestrol

- They are used for HRT in Turner's syndrome, postmenopausal females, contraception (along with progesterone)
- **Side effects:** Breast ca, Uterine ca, Salt & Water retention, Thromboembolism

◆ Selective Estrogen Receptor Modulators (SERM)

Bone (Osteoporosis)			
Breast (Cancer)			
Uterus (Cancer)			
Thrombosis			

- is DOC for ER/PR + breast cancer (premenopausal)
- Raloxifene is used in Px & Rx of postmenopausal osteoporosis
- Toremifene is used for Rx of Breast Ca
- Ospemifene is used for Rx of Post-menopausal dyspareunia

Selective Estrogen Receptor Downregulator (SERD):

- ◆ **Fluvestrant:** its pure anti-estrogenic drug
- Its used for tamoxifen resistant breast cancer.

Palbociclib (Cyclin dependent kinase $\ominus$) + Fluvestrant used for metastatic Hormone receptor (HR) +ve, HER-2 -ve breast cancer

- ◆ **Clomiphene citrate:** MOA: Its estrogen antagonist at anterior pituitary preventing feedback
- Its DOC for Rx of infertility d/t anovulation & oligospermia SE: Multiple gestation
- ◆ **Aromatase $\ominus$:** Aminoglutethemide, Anastrozole, Letrozole, Vorozole, Exemestane
- They are DOC for Rx of postmenopausal

Selective Tissue Estrogen Activity Regulators (STEAR): Tibolone

Use: ↓ Vasomotor symptoms, Vaginal atrophy, Osteoporosis in post-menopausal women

- ◆ **Progesterone**
- Progesterone derivatives: Medroxy, Hydroxy, Dihydroprogesterone
- Testo derivates: Norethisterone, Norgesterel

- Spironolactone derivatives: Drospirenone
- Latest: Desogesterel, Gestodene
- They are used for contraception either with estrogen as combined pill or as minipill, HRT in females, endometrial, breast ca
- Side effects: androgenic effects, salt & fluid retention, irregular menses
- Drospirenone (3rd gen) is used in OCP with estrogen. It has more potent mineralocorticoid effect than spironolactone

Contraception

Combined OCP (Estrogen + Progesterone)

MOA: decreases LH (inhibit ovulation), they have one week of pill-free period

Progesterone only pills (Minipills)

MOA: thickening of cervical mucus & blocking of implantation

Emergency:

<3 days	Levonorgestrel (LNG)	1.5mg single dose / 0.75 mg twice 12h apart	Delays/inhibits ovulation
	YUZPEE regimen (CoCP)	Ethinyl Estradiol (100 mcg) + LNG (500 mcg) twice 12h apart	Delays/inhibits ovulation Prevent fertilization
	SPRM (selective progesterone receptor modulator) Ex: Mifepristone	10-25 mg	Inhibit formation of follicles and endometrial maturation
<5 days	SPRM Ex: Ulipristal	30 mg single dose	



OCP risk (Brest, cervical) risk (Endometrial, Ovarian) Ca.

◆ Mifepristone (Ru-486) : $\ominus$ progesterone receptors

- Mifepristone + Misoprostol is used for abortion. MC side effect is bleeding



MIST

- ◆ **Testosterone:** Testosterone (s.c), Testo-propionate (i.m), Testomesterelone & testo-undecanoate (oral)
 - Its used in Rx of male hypogonadism
 - Nandrolone is used in Rx of aplastic anemia & abused by athletes
 - Side effects: Impotence, Salt & water retention, Oligospermia
- ◆ **Anti-androgens**
 - **5-alpha reductase** ⊖: Fin / Dutasteride used for Rx of BPH, hirsutism & androgenic alopecia
 - **Androgen receptor antagonists:** Flu / Nilu / Apalutamide are used in Rx of prostate ca & hirsutism
 - **Side effects:** Impotence & Gynecomastia (more with androgen receptor blockers)
 - Danazol: ⊖ GnRH so it can be used in endometriosis & gynecomastia
- ◆ **Uterine Stimulants:** Oxytocin (DOC) is used to induce labor & in PPH, Ergometrine in resistant cases
- ◆ **Uterine relaxants:** Ritodrine, Nifedipine, Atosiban are used in premature labor

GH & SOMATOSTATIN DRUGS

- ◆ **GHRH analogue:** Sermorelin is used in Rx of GH deficiency & for DDx of GH deficiency i.e Pituitary vs Hypothalamic cause
- ◆ **GH analogues:** Somatrem & Somatropin
 - Uses: SGA baby, Malabsorption d/t small bowel, AIDS related wasting, Length ↓ (dwarfism)
 - **SE:** Carpal tunnel syndrome, Hyperglycemia, ↑ ICT, Leukemia, Edema

- ◆ **IGF-1 analogues:** Mecasermin
- It is used in Rx of dwarfism by selective IGF-1 deficiency. SE: Hypoglycemia
- ◆ **GH2 receptor antagonist-** Pegvisomant is used in acromegaly
- **Somatostatin analogues:** Oct / Lan / Pasereotide
 - S :** Secretory Diarrhea (Carcinoid Tumor, VIPoma)
 - O :** Oesophageal Varices
 - M :** Malignancy (Glucagon, Insulin, GFR, gastrin "OMA")
 - A :** Acromegaly

STERIODS

Glucocorticoids (GC)	Mineralocorticoid (MC)
Ex: Cortisol (Hydrocortisone) Func: Anti-inflammatory, Immunosuppressant Anti-Cancer Catabolic Carbs (avoid in DM), Protein, Fats (Cushing syndrome), Ca²⁺ (Osteoporosis)	Ex: Aldosterone Func: Na & Water retention Fludrocortisone: Max GC property among MC DOCA: MC with 0 GC property

Short acting (<12 hrs): Hydrocortisone (Max MC property among GC)

Intermediate acting (<36 hrs): Prednisone, Triamcinolone (0 MC property)

Long acting (>36 hrs): Betamethasone (Max potency, 0 MC property)
 Dexamethasone (Max GC property, 0 MC property)

Uses: Anti-inflammatory, Immunosuppressant (Autoimmune dis, Transplantation), Anti-Cancer

Replacement: Addison's dis, Congenital adrenal hyperplasia (to prev virilization)

Fetal lung maturity (surfactant production):

Dexa: 6mg q12h x 4 (total 24 mg in 48 hr)

Beta: 12mg q24 x 2 (total 24 mg in 48 hr)

SE: Glaucoma (Topical use), Cataract (Systemic use)

Osteoporosis, Edema, Infections

C/I in CHF, TB, Renal failure



Autocoids are substances produced by wide variety of cell that act locally at the site of production

Classification:

- Amines : Histamine, Serotonin
- Lipids : Prostaglandins, Leukotrienes
- Peptides : Angiotensin, Bradykinin

HISTAMINE

- ✦ ⊕ H₁: Increase wakefulness

⊕ H₁: Mediates allergy & inflammation, bronchoconstriction,

- ◆ 1st gen H₁ antagonist:

- ◆ Highly sedative- Diphenhydramine, Hydroxyzine,
Dimenhydrinate, Promethazine (Max M ⊖)

Moderate sedative- Pheniramine, Meclizine, Cinnarizine

Mild sedative - Cyclizine, Chlorpheniramine (least sed 1st gen)

- ◆ 2nd gen H₁ antagonist: Cetirizine (derivative of Hydroxyzine),
Levocetirizine, Loratadine (long), Desloratidine (Most potent)
Bilastine, Fexofenadine (least sedative), Astemizole, Terfenadine,
Topical (Alcaftidine, Mizolastine, Olopatidine, Levocarbastine)

➤ USES

H ₁ ⊖: Urticaria, Pruritus, Insect bites
Muscarinic ⊖: Possess - Parkinsonism (EPS) Anti - Acute muscular dystonia Cholinergic - Common cold Property - Px of motion sickness

- Meniere's disease (Vertigo): Betahistine (Vertin)-histamine analogue, Cinnarizine and 1st gen anti histaminics like

- Diphenhydramine, Promethazine are used for Vertigo

- Terfenadine and astemizole have K⁺ channel blocking activity, actively increases the risk of torsades de pointes

SEROTONIN

It acts on 7 types of receptors i.e 5HT1-5HT7. All are G coupled receptors except 5HT3 (ion channel)

➤ 5HT1:

- 5HT1A: decreases anxiety Ex: Bus / Ipsa / Gepirone (5HT1A agonist-Anxiolytics)
- 5HT1B/1D: v.c of cranial vessels Ex: Suma (DOC) / Riza (fastest) / Frova (longest) / Naratriptan (5HT1B/1D agonist)
- ◇ 5HT3: Na/K ion coupled that causes Nausea and Vomitting,
- 5HT3 Antagonists -Dolo / Grani / Ondan (shortest) / Palono (longest & most potent)
- ◇ 5HT4: Increase GI peristalsis
- 5HT4 Agonist -Mosa / Prucalopride used in gastroparesis
- SE: QT-prolongation (Tegaserod, Cisapride)

◇ Migrane management

➤ Acute:

- ✦ 5HT1b/d ⊕: Triptans are C/I in ischemic heart disease, HTN, liver & renal impairment
- ✦ 5HT1f ⊕: Lasmiditan. It has lesser risk of coronary vasospasm
- ✦ Ergot derivatives (non-selective 5HT): Ergotamine, Dihydroergotamine

They can cause gangrene of end arteries especially in limbs, precipitate MI and angina

- ✦ C-GRP receptor $\ominus$: Rime / Ubrogepant
Opioids: Butorphanol, Codein

➤ Prophylaxis:

A: Anti-epileptics (Topiramate, Gabapentin, Valproate), Anti-depressants (Nortriptyline)

B: Beta-blocker - Propranolol (DOC)

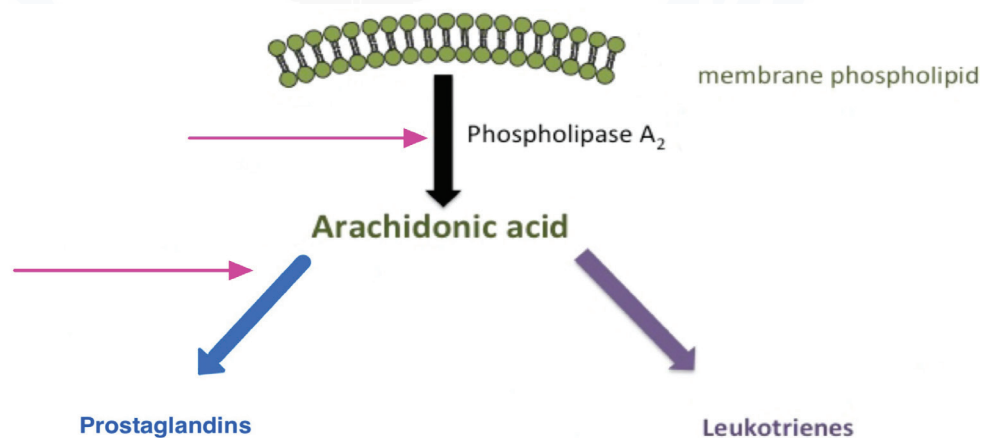
C: Clonidine, CCB (Flunarizine), C-GRP $\ominus$ (Erenumab, Epti / Galca, Fremanezumab)

Of: Onabotulinum

Migraine: Methysergide

Migraine in Preg: PCM (DOC), Sumatriptan

EICOSANOIDS



◆ Prostaglandin E1 analog:

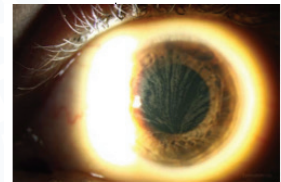
- is used as gastroprotective agent, to maintain the patency of ductus arteriosus, medical abortion & PPH
- Alprostadil is DOC maintain patency of ductus arteriosus & also used for erectile dysfunction

◆ Prostaglandin E2 analog:

- Dinoprostone is used for abortion (2nd trimester), cervical ripening

◆ Prostaglandin F2a analog:

- Carboprost is used for PPH (avoid in asthma), abortion (2nd trimester)
- ✓ Latano / Bimato / Travo / Taflu**prost** is used for glaucoma
- ◆ Nitarsudil+Latanoprost combo can also be used



Cornea verticillata
(vortex keratopathy)



NSAID'S (anti-inflammatory, analgesic and antipyretic properties)

- ✦ Nonselective (COX1=COX2) ⊖:
- ✦ Preferential COX II ⊖: Meloxicam, Nimesulide (hepatotoxic), Etodolac, Ace/Diclofenac, Nabumetone
- ✦ Selective COX II ⊖: Etoricoxib & Lumiracoxib (longest, fastest & most selective), Celecoxib (least selective, shortest)

✦ Non-selective COX ⊖

◆:

- It's used as an antipyretic & analgesic but no anti-inflammatory properties
 - It has additional analgesic effect due to Cannabinoid agonism ⊖ TRPV1 receptor
- Maximum dose of acetaminophen that can be given is 4 g/day.

It causes hepatotoxicity (zonal hepatic necrosis) due to ↑ in NAPQI (toxic metabolite) that ↓ glutathione

- < 1 day : Asymptomatic Liver enzymes
- > 1 day : Emesis plus right upper quadrant pain
- 3-4 day : Liver dysfunction, coagulopathies, metabolic acidosis
- 4 days to 3 weeks : recovery.

- N-acetyl cysteine ⊖ NAPQI and ↑ glutathione

◆ Aspirin:

- It's a salicylate which is non-competitive irreversible ⊖ of COX
- Apart from anti-inflammatory, analgesic and antipyretic properties it also has antiaggregant activity, so it is also used in prophylaxis of MI and stroke

- The anti-inflammatory effect is seen at high dose (325-625 mg) and the antiaggregant effect is seen with lower doses (40-325 mg)
- No antidote is present so forced alkaline diuresis is done to enhance excretion of the drug
- ◆ Indomethacin: DOC for acute gout & used in closure of PDA (DOC: Ibuprofen)
- Sulindac is a prodrug of Indomethacin which can cause Steven Johnson syndrome
- ◆ Ketorolac: It's a potent analgesic providing effect equal to opioids and is rapid and short acting. So it is used in postoperative pain
- ◆ Mefenemic acid: It's very useful in dysmenorrhea

Side effects of NSAIDs:

- GI effects like nausea and vomiting are mc seen. They can also cause Peptic ulcer disease (No GI side effects seen with Selective COX II ⊖)
Cele / Rofe / Valdecoxib are sulfa derivatives so rash & hypersensitivity
- All NSAIDs except low dose aspirin increases risk of MI on chronic therapy
- NSAIDs inhibit antihypertensive effect by inhibiting PG mediated v.d
- Reye's syndrome characterized by liver dysfunction and encephalopathy, so C/I in children

GOUT

☐ Acute Gout:

- **NSAIDS** (except aspirin) are the DOC
- Steroids can be used in NSAIDS resistant cases
- **Colchicine** ⊖ the migration of neutrophils to joints by inhibiting tubulin required for their movement and also decreases inflammatory mediators

☐ Chronic Gout:

- ◆ **Xanthine Oxidase** ⊖ - Allopurinol (DOC), Febuxostat (If allopurinol allergy)

Allopurinol is also used in Lesch-Nyhan syndrome and Tumor lysis syndrome

Side effects- Xanthine stones can form in the urine

Dose of 6-MP and azathioprine should be decreased when used with allopurinol

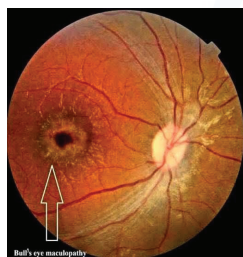
- ◆ **Uricosuric drugs:** Probenecid, Sulfapyrazone, Benbromarone, Lesinurad

Probenecid increases uric acid excretion so C/I in patients with renal stones

Losartan also has uricosuric effects

Urate $\longrightarrow$ Allantoin

- ◆ **Uricase analogs:** Rasburicase (DOC), Pegloticase is used Rx of tumor lysis syndrome



Active space

RHEUMATOID ARTHRITIS

- ❑ **Acute attack:**
NSAIDS (Aspirin, Ace/diclofenac, Celecoxib) and
Steroids (Triamcinolone, Prednisolone)
- ❑ **Chronic:** DMARDS (Disease modifying Anti-Rheumatoid drugs) /
SAARDS (Slow acting Anti-Rheumatoid drugs)
- ❑ **Conventional DMARDS**

- ✦ **Methotrexate:** It's the DOC and mostly the treatment is started with MTX later HCQS is added
- ✦ **Sulfasalazine** an Amino-salicylic acid derivative can be used (also used in UC)
- ✦ **Hydroxychloroquine** SE: Bull's eye maculopathy

◇ **Biologicals**

- ✦ **IL-1 ⊖** : Anakinra
- ✦ **IL-6 ⊖** : Tocilizumab, Sarilumab
- ✦ **Anti-TNF:** Infliximab, Certolizumab, Adalimumab, Golimumab, Etanercept SE: infections

Other uses: Psoriasis, IBD, Ankylosing spondylitis, Rheumatoid arthritis (PIAR) Mnemonic: MARD Pyaar mei Khaid (CAGE) hota hai

- ✦ **JAK ⊖** : Bari, Upada, Tofacitinib
(Baricitinib approved alopecia areata)
- ✦ **CD-20 ⊖** : Rituximab CD80/86 ⊖ : Abatacept
- ✦ **Immunomodulators:** cyclosporine and azathioprine
- ✦ **Leflunomide**



- ❑ **Pharmacokinetics:** Effects of body on the drug
- ❑ **Pharmacodynamics:** Effect of drug on the body
- ✦ Pharmacokinetics includes ADME (Absorption, Distribution, Metabolism, Excretion)

Routes of Administration:

- 1) **Local** - Topical (Ointment>Gel>lotion), Intra-articular (steroids in RA), Intrathecal (MTX in childhood leukemia, Baclofen),
- 2) **Systemic** - Enteral (Oral, Rectal), Parenteral

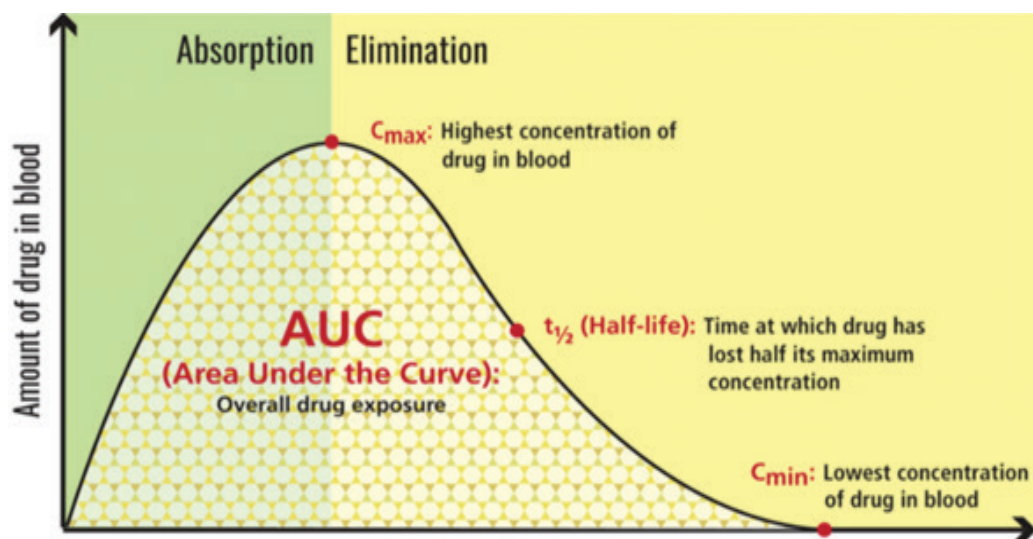
- ◆ **ABSORPTION:** Movement of drug from site of administration to blood

Lipid Soluble drugs (Non - ionized or uncharged drug) cross the membrane easily

Water soluble drugs (Ionized or charged drug) cross the membrane

- Whenever the medium is same the drug will cross membranes (Acidic drugs in acidic media, Basic drugs in basic media)
- is the pH at which the drug is 50% ionized & 50 % non-ionized. If $\text{pH} - \text{pK}_a = 1$ (one form is 90% & other is 10%), for difference of 2 (one form is 99% & other is 1%), for difference of 3 (99.9% & 0.1%)

- **Bio-availability:** % of the given dose which reaches the systemic circulation in unchanged form
- Bio availability of i/v route - 100%

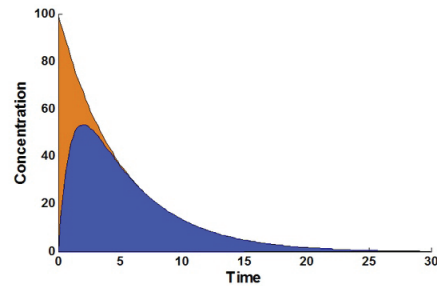
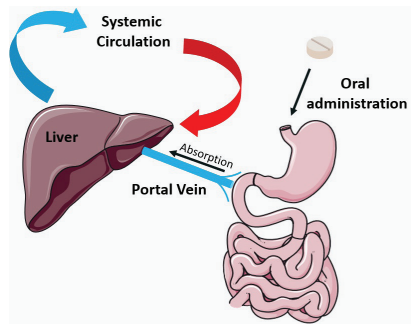


Factors:

- Absorption** - more absorption, more bio availability
- Metabolism** - 1st pass metabolism or pre - systemic metabolism
 Drugs with high first pass metabolism
 (Nitrates, Have-Hydrocortisone, Large-Lignocaine,
 Pre-Propanolol, Systemic-Salbutamol, Metabolism-Morphine)

Oral route has maximum 1st pass metabolism. No 1st pass metabolism, bio availability is more

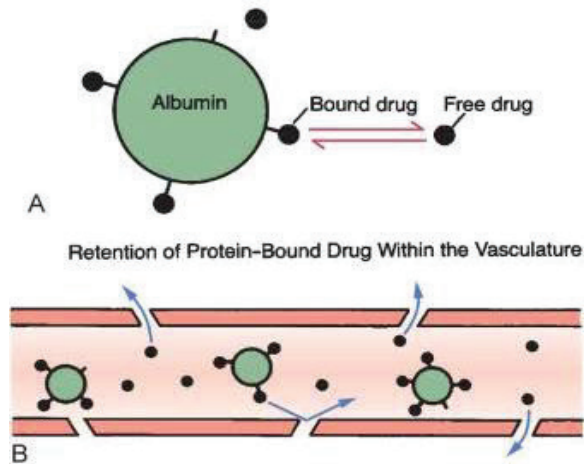
Ex: sublingual NTG Advantage: 1) No 1st pass metabolism 2) Used in emergency 3) After getting the desired action the rest of the drug can be spitted, 4) Self administration



- ◆ **Distribution:** After a drug enters the blood it may be distributed to various tissues
- If more amount of drug is entering the tissue it has more volume of distribution and vice versa (More LS drug - More distribution)
- More the plasma protein binding less the distribution
- ✓ Plasma protein binding (PPB): Free form of drug is in active form
- 1) **Q. Acidic drugs binds to Albumin** (Barbs, Benzo, Phenytoin, Valproate, Penicillins, Cephalosporins, Tetracyclines, Warfarin, Sulfonamides, NSAIDS)
- 2) **Basic drugs binds to α1 - Acid Glycoprotein** (Pilocarpine, Physostigmine, Atropine, Amphetamines, Verapamil, B-blockers, Lidocaine, Quinidine, TCA, Opioids)

Ex: Sulfonamides if given along with warfarin, there is competition for albumin as both are acidic. Warfarin gets displaced resulting in toxicity, so they are not given together

- Ex: Sulfonamides if given along with warfarin, there is competition for albumin as both are acidic. Warfarin gets displaced resulting in toxicity, so they are not given together



- ❑ P-Gp (efflux transporter): important for digoxin toxicity^Q and anti-cancer drug resistance

P-Gp $\ominus$ ($\uparrow$ toxicity): Verapamil, Amiodarone, Quinidine, Clarithromycin, Itraconazole, Navir (Ritonavir), Erythromycin (Mnemonic: Vagcine),
 P-Gp inducers are Enz inducers like Rifampicin

Teratogenic drug:

- The drugs given during pregnancy which cross the placenta & cause damage of fetus
 Eg: Thalidomide - was used for morning sickness in pregnancy earlier, now used for multiple myeloma and type II lepra reaction
 S/E - ϕ phocomelia

✓ **Volume of distribution** ($V_d = \dots\dots\dots$)

- If V_d is $\uparrow$ - distribution is $\uparrow$
 Clinical importance of V_d :



Active space

- Loading dose required to attain a particular plasma concentration depends on V_d
 $LD = \dots\dots\dots$

➤ Maintenance dose: Amount of drug required to maintain the target plasma concentration $MD = \text{clearance} \times \text{Target plasma concentration}$

➤ Dialysis is effective for drugs having low V_d and low PBP. Its not effective in (Amphetamines, Verapamil, Op & opioids, Imipramine, Digitalis, Dialysis - Diazepam)

Redistribution is seen with Thiopentone

◆ Elimination: Stoppage of action of drug

1) Metabolism - breaking down

2) Excretion - Removing the drug from the body

➤ Metabolism and excretion is together called as elimination

➤ Most of the drugs are inactivated by metabolism but some may be activated from inactive drug (pro drugs) Ex: Levodopa metabolism activates it into Dopamine

Prodrugs:

All - All ACEI, ARBS - except capto / & Lisinopril, Olme /

Candesartan,

Pros - Proguanil,

Prefer - Prednisone, PPI,

Doing - Dipevefrine,

Masters - 6-MP, 5-FU, Minoxidil, Methyldopa & Levodopa,

In - Irinotecan,

Clinical - Cyclophosphamide, Clopidogrel, Carbimazole,

Subjects - Statins like lovastatin, simvastatin, Sulfasalazine, Sulindac

➤ **Metabolism** - make a drug water soluble

PHASE - 1 reaction	PHASE - 2 reaction aka conjugation reaction
Oxidation(MC)	Glucoronidation (MC)
Reduction	Acetylation
Hydrolysis	Methylation
Cyclization	Sulfation
Deamination	Glutathionation

- **Phase 1** - Enzymes are present in E.R (microsomal) that attach the functional group to the drug
- **Phase 2** - These reactions take place in cytoplasm (nonmicrosomal) except Glucoronidation that takes place in E.R (Microsomal). Every drug become water soluble after phase 2 metabolism

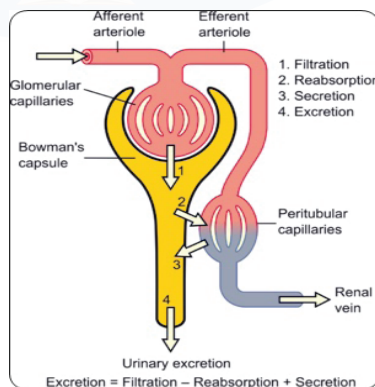
Aspirin toxicity- acidic drug so makes urine alkaline (basic) by NaHCO_3 so all the drug can be excreted completely

Amphetamine toxicity- basic drug so make the urine acidic by NH_4Cl

Cytochrome P450

- Cytochrome 3A4 are responsible for metabolism of most drugs followed by CYP2D6 (Codein to morphine, most CNS & CVS drugs). CYP2C9 (Warfarin), CYP2C19 (Clopidogrel, Omeprazole), CYP2E1 (PCM maked NAPQI), CYP1A (Theophylline)
- **Acetylation** Ex: Drugs metabolized by acetylation are Dapsone, INH, Nitrazepam, Clonazepam, Hydralazine, Acebutalol, Kaffeine, Sulfonamides, Procainamide (DINCHAK Sexy Pooja) These drugs cause drug induced SLE in slow acetylators

- **Enzyme Inducers:** St. John's wort, Carbamazepine, Cigarette smoke, Rifampicin, Alcohol (chronic), Phenytoin, Phenobarbital, Griseofulvin (So CRAP)
They cause drug failure Ex:
- **Enzyme Inhibitor:** Valproate, INH, Protease $\ominus$, Cimetidine, Ciprofloxacin, Omeprazole, Ketoconazole, Erythromycin, Grapefruit juice, Quinidine (VIP COKE & Grapefruit juice)
They cause drug toxicity Ex: Erythromycin causing Theophylline toxicity
- **Excretion:** Removal of drug from the body
 - Function of Glomerulus: 1) Filtration, 2) Tubular reabsorption, 3) Tubular secretion
 - 1. **Filtration:** Depends on PBP (protein bound drugs can't be filtered) and renal blood flow but not on solubility of drugs (Lipid and water-soluble drugs both are filtered)
 - 2. **Tubular reabsorption** - More the lipid solubility of the drug, more the reabsorption and less excretion. Urine's pH can be changed to make drugs ionized by Sodium bicarbonate or Ammonium chloride in toxicities
 - 3. **Tubular secretion:** doesn't depend on lipid solubility or PBP
- 2 types of pumps present in proximal tubules: one is for acidic and other is for basic drug. Only one drug can pass at a time Ex: Probenecid is given together with penicillin. They are competitors for tubular secretion. Probenecid is excreted, penicillin remains in the body



- **RATE OF ELIMINATION:** Amount of drug eliminated divided by time taken
 $R = \text{Amount of drug eliminated} / \text{Time taken}$
- **CLEARANCE (CL):**
 $CL = \text{Rate of elimination} / \text{Plasma conc} (CL = UV/P)$
- **HALF LIFE ($T_{1/2}$):** It is time in which plasma concentration becomes 50%
 $T_{1/2} = 0.693 \times V_d / CL$
- It takes 4-5 half lives to remove most of the drug out of the body and to achieve steady state concentration

ORDER OF PHARMACOKINETICS

First order ($T_{1/2}$ constant)	ZERO ORDER ($T_{1/2}$ variable)
Fraction eliminated per unit time is constant Ex: 500 mg of a drug removed at 50% /hour	Amount is constant Ex: 500 mg drug removed at the rate of 50 mg/hour
500 - 250 - 125 - 62.5 - 31.25 - 15 - 7.5	500 - 450 - 400 - 350 - 300 - 250 - 200 - 150
Around 95% is cleared in 4-5 half lives	
Rate of elimination $\propto$ Plasma Conc $\uparrow$ concentration - $\uparrow$ Rate of elimination	Rate of elimination is constant or independent of plasma concentration
Clearance is constant	Clearance is inversely proportional to plasma concentration

Zero order kinetics drugs:

W - Warfarin,

A- Aspirin / Alcohol

T - Theophylline,

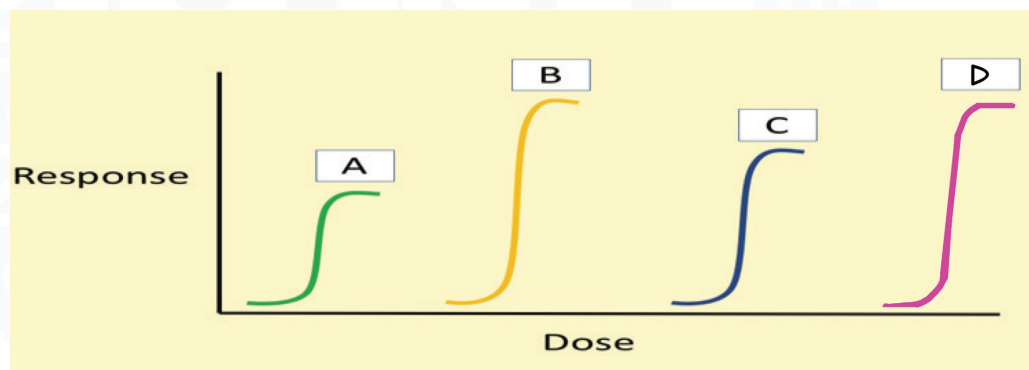
T- Tolbutamide (Anti diabetic),

POWER - Phenytoin

Patients are more prone to have toxicity with zero order drug

PHARMACODYNAMICS

- This is what the drug does to the body to produce an effect by binding to a target like receptor, enzymes, ion channels, transporters
- **Potency:** ability of a drug to bind to the target. Its related to dose of the curve
- **Efficacy:** ability of a drug to produce an effect after binding to the target
- In DRC (dose response curve) X axis is potency(dose) and Y axis (height) is efficacy



Max efficacy

Max potency

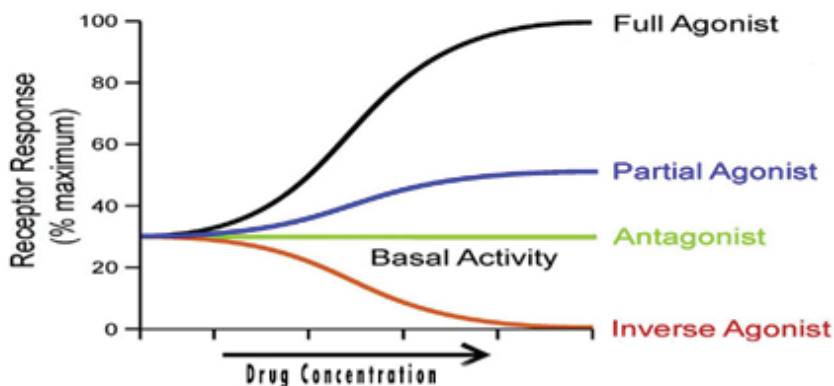
Least efficacy

Least potency

- In children the dose is roughly $\frac{1}{3}^{\text{rd}}$ of adult dose

- 1) **Agonist** - Drugs produce maximal activity with intrinsic efficacy +1
- 2) **Partial agonist** - Drugs produce sub maximal activity with intrinsic efficacy 0 to +1

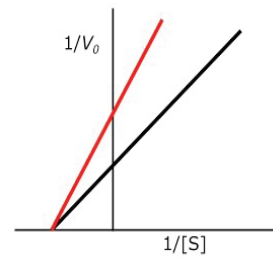
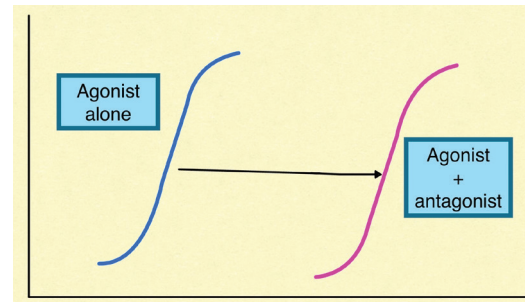
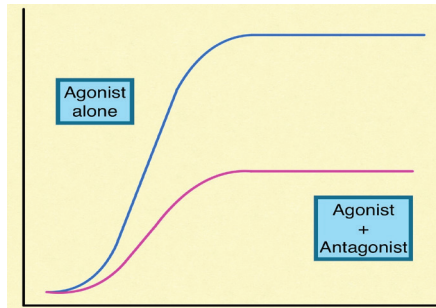
- 3) **Inverse agonist** - drugs produce opposite action with intrinsic efficacy 0 to -1
- 4) **Antagonist** - bind to the receptor but no action with intrinsic efficacy 0, It itself has no action but stop the action of the other drug



TYPES OF ANTAGONISM

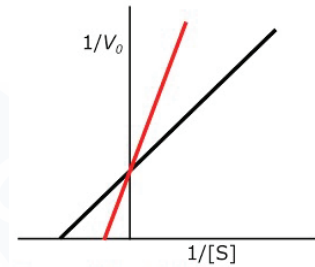
◆ Pharmacological antagonism

- ◆ **Competitive (same receptor)** - causes parallel shift to the right in DRC for agonists. It ↑ the potency of the drug and can be reversed by ↑ the dose of the agonist (↑ K_m but V_{max} remains same)
 - Competitive reversible - MC type of antagonism Ex: B-blockers blocking the effect of Catecholamines on B1 receptors
 - Competitive irreversible - Rare type Ex: Aspirin inhibiting TXA₂, PPI inhibiting proton pumps, Phenoxybenzamine inhibiting receptors. DRC looks like non-competitive antagonism
- ◆ **Non-competitive:** One drug prevents another receptor action by binding on a site distal to the receptor of agonist. It decreases the efficacy (V_{max}) of the agonist



Noncompetitive Inhibition

Lineweaver - Burk Plot



Competitive Inhibition

◆ Physiological antagonism

One drug prevents another drug's action by binding to another receptor. Ex: B.C caused by H1 receptor prevented Adrenaline binding to B2 receptors, TXA2 prevented by PGI2

◆ Chemical antagonism

One drug prevents another drug's action by chemically combining in a solution. Ex: Protamine sulphate to Heparin, MAB to protein targets, Dimercaprol to heavy metals

◆ Pharmacokinetic Antagonism

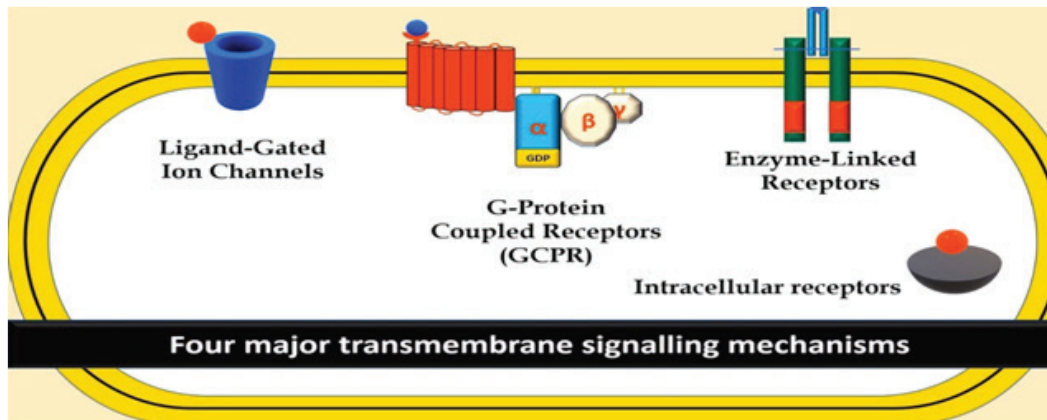
One drug prevents another drug's action by acting on its pharmacokinetic properties

Ex: Enzyme inducers decrease the effect of other drugs by increasing their metabolism

Charcoal binds to alcohol preventing its absorption

Probenecid ⊖ excretion of Pen-G

Signal transduction mechanisms



1) Ligand gated ion channel receptor

- Ligand action on ion channel receptor open ion channels leading to depolarization or hyperpolarization quickly so these are the fastest acting receptors
- Ex: GABA-A receptors, Glutamate receptors - NMDA, AMPA, Ach-Nicotinic receptors, 5HT3

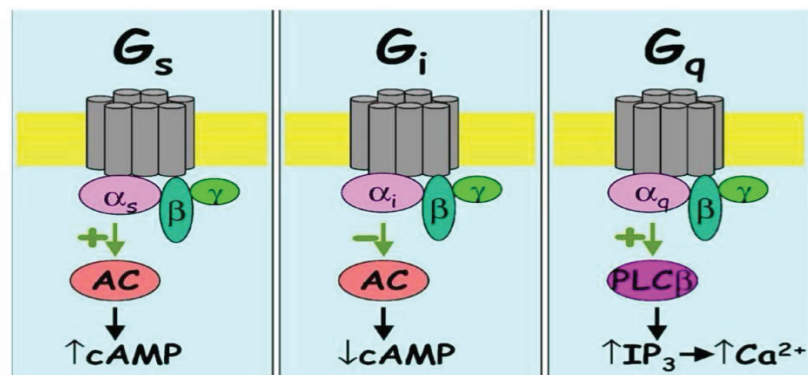
2) Enzyme linked receptors

- Activation of extracellular receptor is linked to intracellular enzymes like tyrosine kinase, janus-kinase etc which produces intracellular effects
- Ex: Tyrosine kinase - Insulin, EGFR, TLR, VEGFR)

Janus kinase receptors - GH, Prolactin, Cytokines,
Serine / threonine receptors (TGFR)

3) G - Protein Coupled Receptors (GPCR)

- G-protein coupled receptors are located outside cell membrane & twist 7 times while entering cytoplasm so aka 7 transmembrane /heptahelical/serpentine receptors
- **G_s**: Its activation, stimulates adenylyl cyclase that ↑ C-AMP which inturn stimulates Protein kinase A that phosphorylates proteins & calcium channels Ex: B1, B2, D1
- B1 phosphorylates calcium channels ↑ intracellular calcium contracting the myocardium
- B2 phosphorylates protein in smooth muscle i.e MLCP that dephosphorylates myosin relaxing smooth muscle
- **G_i**: Its activation, ↓ CAMP that opens K channels Ex: M2, μ , α_2 , D2, 5HT1
Ex: M2 ↓ CAMP, ↓ calcium that ↓ contraction, HR, conduction
- **G_q**: Its activation stimulates Phospholipase C that hydrolyses PIP2 (phosphoinositol bisphosphonate) into IP-3 (inositol triphosphate that increases calcium) & DAG (di acyl glycerol that stimulates Protein Kinase C to phosphorylate ion channels, receptors etc) Ex: M1, M3, α_1 , AT1&2, V1



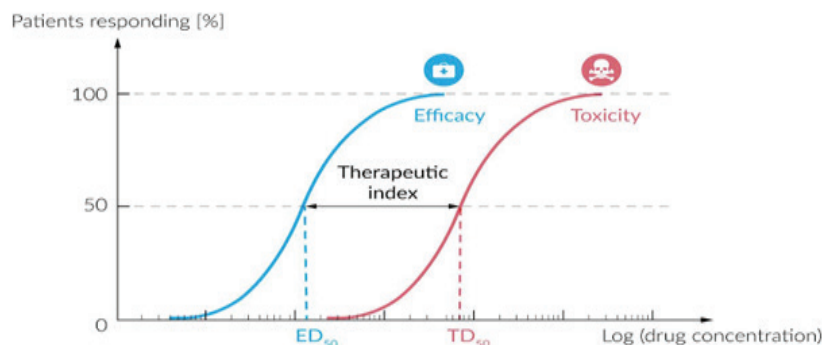
4) Intracellular receptors (Nuclear receptor superfamily)

- Only lipid soluble drugs can act on intracellular receptors. Although they are of 2 types i.e Nuclear & Cytoplasmic they are still called as Nuclear receptors since cytoplasmic receptors also translocate to nucleus & regulate gene expression.
- If drug binds with nuclear receptor then they bind with DNA and acts on replication, transcription & translation so these processes take time. That's why these are the slowest acting receptors
- **Cytoplasmic receptor** Ex -Corticosteroids (Glucocorticoid, Mineralocorticoid receptors), Vit-D
- **Nuclear receptor** Ex - PPAR, Sex hormones, Vit-A, Thyroid hormones (PSVT)

THERAPEUTIC INDEX, WINDOW

◆ Therapeutic Index:

- It's a measure of drug safety calculated as $Q.T.I = LD_{50} / ED_{50}$ (Mnemonic: TILE)
- **LD 50:** minimum dose required to kill 50 % of population (measure of drug toxicity)
- **ED50:** minimum dose required to produce clinical effect in 50 % population (measure of drug potency)



- For the drugs with Low T.I, therapeutic drug monitoring (TDM) should be done

A: Anti-epileptics, Aminoglycosides, Anti-arrythmics

LoW: Lithium

Therapeutic: Theophylline,

Drug: Digoxin

Causes: Cyclosporin, Tacrolimus (Immunosuppressants)

Toxicity: TCA

Therapeutic Window

It's the range of plasma concentration ranging from minimum effective concentration (MEC) to minimum toxic concentration (MTC) i.e the drug produces adequate clinical effect without toxicity within this range. This is a better measure when compared to T.I

CLINICAL TRIALS FOR NEW DRUG DEVELOPMENT

Phase 0 : Microdosing (100 mcg)
to determine PK & PD Size: 1-5

Phase 1: Human safety & Pharmacology trial
Drug tested on normal human volunteers (efficacy can't be measured as they aren't patients). Mainly to test toxicity (max tolerable dose) and pharmacokinetics of the drug Size: 10-100

Phase 2: Therapeutic exploratory trial
Drug tested in small number of patients. This phase is to establish both efficacy and safety in patients Size: 100-500

Phase 3: Therapeutic confirmatory trial
Drug tested in large group of patients with different genetic & ethnic backgrounds to confirm efficacy. If safe application is filed with FDA /CDSCO (Central Drugs Standard Control Org) and if approved the drug is marketed Size: 1000-5000

Phase 4: Post marketing surveillance
To know rare or long-term use adverse effects

Phase 5: Pharmacoepidemiology
To check new indications of drugs and confirm SE found in Phase 4

Pharmacovigilance It is the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other medicine/vaccine related problem.

Type A - related to mechanism of drug Ex: vasodilators leading to headache, hypotension

Type B - not related to MOA of drug

Type D - teratogenicity

Type E - adverse effect - apart from drugs that show dependence, some drugs show withdrawal reaction on sudden cessation of drug. Ex: Suddenly stopping Anti-epileptics precipitate seizures

Schedule of drugs

✓ Schedule H - only available on prescription by registered medical practitioner

✓ Schedule H1 - antibiotics

✓ Schedule G - given only under medical supervision

✓ Schedule X - Narcotic and Psychotic drugs (NRx) Ex: Morphine, Opioid, Benzodiazepine, Barbiturates

✓ Schedule Y - NDCTR (New drugs & clinical trial rules)

Pregnancy category

Category A>B>C : safe in pregnancy

Category D: Benefit > teratogenic risk

Category X: Teratogenic risk > Benefit (absolutely contraindicated)

Ex: Warfarin (Contradi syndrome), Misoprostol (Moebius syndrome), Diethylstilbestrol (Vaginal adenocarcinoma)

Correct format of prescription: Complete name of drug, dose, frequency (no abbreviations)

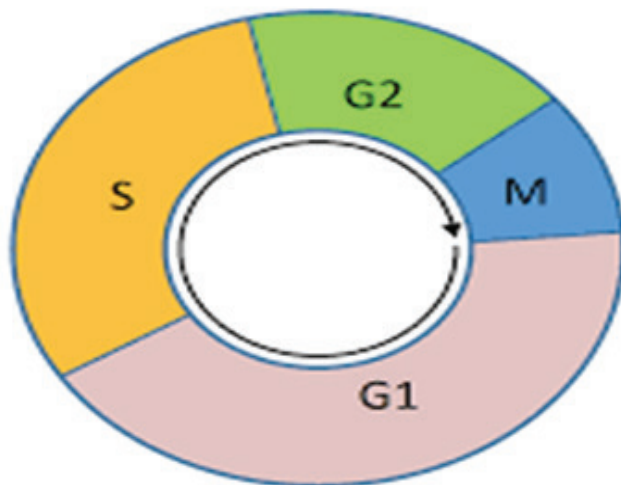
Ex: Tablet azithromycin 500 mg once a day for 5 days

Expiry date of a drug - decreased efficacy, company wont be responsible for any adverse effect arising from expired drug

Type of Drug	Description	Example
Spurious / Counterfeit	Faxe; missing or wrong active ingredient	"Paracetamol" tablet with no PCM
Misbranded	Wrong label / low quality	PCM 650 label but only PCM 300 content
Adulterated	Contains unwanted or harmful impurities	Contaminated drug
Orphan	For rare or neglected diseases	



- Anti-neoplastic drugs commonly damage rapidly dividing cells i.e tumor cells & also normal cells
- General side effects include Nausea Vomiting [Rx: 5HT3 blockers (Ondan / Palonosetron) or NK1 blockers (Aprepitant)], mucositis, alopecia



Classification

- ◆ Non-cell cycle specific: Alkylating agents, Platinum compounds, Anti-tumor antibiotics
- ◆ Cell cycle specific:
 - ◆ **S-phase specific** - Antimetabolites (Anti-folate, Purine analogs, Pyrimidine analogs), Topoisomerase $\ominus$, Histone deacetylase $\ominus$, Hydroxyurea
 - ◆ **M-phase specific** - Vinka alkaloids, Taxanes

Alkylating Agents (AA):

- A) Nitrogen mustards → Cyclophosphamide*, Ifosfamide, Melphalan*, Chlorambucil, Mechlorethamine
- B) Busulfan
- C) Nitrosoureas → Carmustine, Lomustine, Streptozocin
- D) AA by methylation → Procarbazine, Dacarbazine, Temozolomide

◈ Cyclophosphamide

Side effects:

- ❑ It's degradation product is acrolein produces Hemorrhagic Cystitis (Rx: MESNA)

SIADH, Cardiotoxicity, Bone marrow Suppression

Platinum compounds (PC): Cis / Carbo / Oxaliplatin

- MOA: AA & PC bind to DNA at N7 position of guanine which leads to mispairing eventually leading to apoptosis. Although they are non-cell cycle specific most action is seen on G1-S phase

- ◈ Cisplatin - it's the most-wide spectrum anti-cancer drug

Side effects:

- ❑ Most emetogenic drug
- ❑ Nephrotoxic (prevented by chloride diuresis),
Otototoxic : Max with cisplatin, least with Carboplatin
Neurotoxic : Max with carboplatin
- ❑ Nephrotoxicity of cisplatin is decreased by using Amifostine

Anti-tumor antibiotics:

Anthracyclins (Dauno, Doxo, Epi, Ida)

Bleomycin, Actinomycin D (Dactinomycin), Mitomycin C

- MOA: Inhibit topoisomerase II in DNA creating free radicals causing DNA damage in all phases especially G2 phase
- Actinomycin D binds to DNA & Drug-DNA complex inhibit RNA. So it works on DNA & RNA
- Side effects are due to free radicals that damage heart (cardiomyopathy), lungs (pulmonary fibrosis)

◆ Anthracyclins: Doxorubicin (Adriamycin), Daunorubicin

Side effects

- ❑ Cardiotoxicity: Dilated cardiomyopathy due to free radicals
antidote is Dexrazoxane
- ❑ Red urine - Doxo & Daunorubicin SE : Radiation recall syndrome
- Bleomycin: Side effect: Pulmonary fibrosis, Flagellated dermatitis
- Actinomycin- D (Dactinomycin): binds to DNA inhibiting RNA polymerase Side effect is Radiation recall syndrome
- Mitomycin C: Intra vesical for Bladder ca, Topical for laryngotracheal (subglottic stenosis)

Anti-metabolites: They act in the S-phase of cell cycle

a. Folic acid analogues: Methotrexate, Permetrexed (used in Lung ca)

➤ **MOA:** $\ominus$ Dihydrofolate reductase

Methotrexate uses: Mnemonic **CANCER**

Chron's disease, Abortion, Non-Hodgkins lymphoma,
Choriocarcinoma, Ectopic Pregnancy, Rheumatoid arthritis

Side-effects:

Bone marrow suppression and Mucositis d/t DNA synthesis inhibition (Rx - Leucovorin rescue which is folinic acid),
Hepatotoxicity

b. Purine analogues: 6-mp (Mercaptopurine) (Azathioprine (analogue of 6-MP), 6-TG (Thioguanine), Pentostatin, Cladribine

MOA: inhibit HGPRT that $\ominus$ de-novo purine synthesis and incorporated into DNA (abnormal DNA)

c. Pyrimidine analogues: 5- FU (Fluro-uracil) Capecitabine (prodrug of 5-FU), Gemcitabine, Cytarabine

SE: Bone marrow suppression, Hand foot syndrome (also from Capecitabine), Pulmonary toxicity

Mitotic spindle $\ominus$:

I. Vinca alkaloids: Vincristine, Vinblastine, Vinorelbine

II. Taxanes - Pacli, Docetaxel

Vincristine causes neurotoxicity, SIADH

Vinblastine causes bone marrow suppression

Topoisomerase $\ominus$: **MOA:** DNA replication forms multiple coils which creates tortional stress. DNA is religated by topoisomerase I (inhibited by Irinotecan & Topotecan) topoisomerase II (Etoposide & Tenoposide)

◆ Tyrosine Kinase $\ominus$:

- ◆ CML: Imatinib, Nilotinib, Dasatinib (BCR-ABL)
- ◆ GIST: Sunitinib, Imatinib, Regorafenib
- ◆ Lung cancer: Ceritinib, Afatinib, Gefitinib, Erlotinib
- ◆ Breast cancer: Lapatinib ($\ominus$ HER -2)
- ◆ HCC: Sorafenib
- ◆ RCC: Pazopanib, Axitinib, Sorafenib, Sunitinib
- ◆ Medullary ca of thyroid: Vandetinib, Lenvatinib

◆ Human Epidermal growth factor receptor (HER)

- ◆ HER -1 $\ominus$: Cetuximab, Panitumumab, Necitumumab,
- ◆ HER-2 $\ominus$: Trastuzumab, Pertuzumab
- ◆ Anti-Veaf : Ramucirumab

Drugs/ TOC

Hemat Ca

- ◆ AML: Dauno/Idarubicin + Cytarabine (Ara-C)
- ◆ CML: Imatinib
- ◆ ALL: Vincristine + Prednisolone + Asparaginase + Daunorubicin (VPAD)
- ◆ CLL: Fludarabine + Cyclophosphamide + Rituximab (FCR)
- ◆ Hodgkin's: Adriamycin, Bleomycin, Vinblastine, Dacarbazine (ABVD)
- ◆ Non-Hodgkin's: Cyclophosphamide + Hydroxydaunorubicin + Oncovin (Vincristine) + Prednisolone + Rituximab (CHOP-Regimen) or FCR regimen

Brain tumors - DOC Temozolomide

Head & Neck Ca - Cisplatin + 5FU

Thyroid cancer: DOC Iodine 131/ Sorafenib

✓ Medullary Ca of thyroid - Vandetinib

Lung cancer - Cisplatin

Breast Ca:

✓ Pre-menopausal: Tamoxifen,

✓ Post Menopausal: Aromatase inhibitor (Letrozole)

GI tumors - DOC Upper (Esophageal, Gastric) - Cisplatin+ 5FU,

✓ Colorectal: FOLFOX (folinic acid + 5FU + oxaliplatin)/FOLFIRI (folinic acid + 5FU+ Irinotecan)

✓ Anal - 5FU + Mitomycin C

✓ Pancreatic: Gemcitabine

Kidney - Sunitinib/ Sorafenib, Pembrolizumab

HCC - Sorafenib

GU tumors - DOC Cisplatin

✦ Bladder: Cisplatin + Gemcitabine

✦ Ovarian & Cervical: Cisplatin + Paclitaxel

✦ Testicular: BEP (Bleomycin + Etoposide + Platinum compound i.e Cisplatin)

Osteosarcoma: DOC - Cisplatin



Cisplatin/Carboplatin → acoustic nerve damage (and nephrotoxicity)

Vincristine → peripheral neuropathy

Bleomycin, Busulfan → pulmonary fibrosis

Doxorubicin → cardiotoxicity

Trastuzumab → cardiotoxicity

Cisplatin / Carboplatin → nephrotoxic (and acoustic nerve damage)

Cyclophosphamide → hemorrhagic cystitis

5-FU → myelosuppression

6-MP → myelosuppression

Methotrexate → myelosuppression

Radiosensitizer drugs: Actinomycin-D Bleomycin, Cisplatin, Doxorubicin, 5-FU, Gemcitabine, Hydroxyurea



Active space

CELL WALL ⊖

- a. **Glycopeptides:** Orita / Dalba / Tela **vancin**, Vancomycin, Teicoplanin
 Vancomycin is effective for a+ve and its DOC for MRSA & Enterococcus. Its also used for Clostridium difficile
 its not absorbed orally
 VRSA develops due to change of d Alanine-Alanine to d Alanine-Lactate
 Side effect: Red man syndrome
- b. **Beta-lactams:** MOA: inhibit cross linking of peptidoglycan chain of the cell wall by inhibiting the enzyme transpeptidase

Penicillins: Gram +ve and Gram -ve

- Penicillin G (acid labile): DOC for Gas gangrene, Meningococcus, Actinomyces, Staph & Strep, Tetanus, Treponema, Anthrax, N. meningitis, Leptospira, Yaws (G MASTAN Loves Yaw)

Acid resistant Penicillins: Pen V, Oxa / Dicloxa / Cloxa / Amoxi / Ampicillin (cillin)

- Extended spectrum penicillins: Aminopenicillins (Ampicillin & Amoxicillin), Carboxypenicillins (Carbenicillin, Ticarcillin) and Ureidopenicillins (Piperacillin, Azlocillin, Mezlocillin)

- Aminopenicillins: active against both Gram +ve and -ve organisms

Ampicillin is DOC in Listeria and Enterococcus faecalis

- Carboxy and Ureidopenicillins: active against Gram -ve including pseudomonas and Enterobacter Klebsiella

Side effects of Penicillins:

- ❑ Hypersensitivity in the form of rash is the MC (especially ampicillin in EBV) and most severe is anaphylaxis
- ❑ If a person develops severe allergic response to penicillins all other B - lactams should be avoided except monobactam (aztreonam)
- ❑ Oral penicillins causes NVD, Pseudomembranous colitis, Methicillin causes interstitial nephritis

Resistance to Penicillins:

- Beta lactamase production
- ESBL (Extended spectrum Beta-lactamase): Carbapenems are DOC
- B-Lactamase $\ominus$ and Drug combos

o Clavulanic acid + Amoxicillin / Ticarcillin
 o Sulbactam + Ampicillin / Cefoperazone
 o Tazobactam + Piperacillin / Cefoperazone
 o Avibactam + Ceftazidime
 o Relebactam + Imipenem
 o Vaborbactam + Meropenem

- Alteration in Penicillin binding protein (PBP) especially in Staph Aureus which is plasmid mediated

◆ Cephalosporins: Gram +ve and -ve

- 1st Generation (Gram +ve, Anaerobes, Gram -ve): Cephalexin, Cephalothin, Cefazolin, Cephaloridine, Cefadroxil
- Cefazolin is DOC for surgical prophylaxis

- **2nd Generation** (Gram -ve, Anaerobes): Cefaclor, Cefotetan, Cefamandole, Cefprozil, Loracarbef, Cefoxitin, Cefuroxime (has good CSF penetration among 2nd generation)
- **3rd Generation** (Gram-ve, Gram +ve): Cefixime, Cefpodoxime, Cefotaxime, Ceftriaxone, Cefoperazone, Ceftazidime, Ceftibuten, Cefdinir, Cefditoren, Cefpiramide, Moxalactam
 - All of them cross BBB except Cefoperazone and Cefixime
 - Ceftazidime and Cefoperazone are active against pseudomonas. Ceftazidime is the DOC for Rx of pseudomonas and meloidosis
 - Ceftriaxone is DOC for Bacterial meningitis (empirical), Hemophilus influenza, Typhoid, Proteus, E.coli, Gonorrhea (Big, Heavy, Tight PEG)
- **4th Generation**: Cefepime, Cefpirome
- **5th Generation**: Ceftaroline, Ceftobiprole (Use : MRSA)
- **6th Generation**: Cefiderocol
 - Its structurally similar to Ceftazidime & Cefepime. It binds to Fe³⁺ and uses bacterial ion transport system to enter periplasmic space (Trojan horse).
 - It has activity against drug resistant bacteria including Carbapenem & Beta lactamase resistant etc

Cefepime + Enmetazobactan / Cefiderocol - Rx of Complicated UTI

With each generation the spectrum and resistance to beta-lactamase increases

Cephs are not effective against Listeria, Atypical (mycoplasma), MRSA, E.fecalis (LAME)

Side effects:

- Hypersensitivity like Penicillins
- Nephrotoxicity with most cephalosporins except Cefoperazone and Ceftriaxone which are excreted in bile (Safe in renal failure, cause gallstones)
- Cefamandole, Cefotetan, Cefoperazone, Moxalactam can cause disulfuram like reactions and thrombocytopenia

◆ **Carbapenems:** Gram +ve, Gram -ve, Anaerobic

- They are the most-wide spectrum drugs like Imi, Mero, Ert, Dori, Faropenem
- Imipenem is metabolized in kidney by dehydropeptidase producing nephrotoxic metabolites so it's always combined with cilastatin which is dihydropetidase $\ominus$
- They are DOC for Klebsiella and active against ESBL producing organisms like Enterobacter, Serratia. They are also active against Pseudomonas, Acinetobacter

SE: Hypersensitivity and seizures (Imipenem and Meropenem)

◆ **Monobactam:** Gram -ve only

- Aztreonam is the only B-Lactam that can be used in patients with allergies to penicillin

Fosfomycin: It's an analogue of PEP and $\ominus$ NAM synthesis used for UTI's in pregnant women

SAFE IN RENAL DISEASE

Safe (ceph): ceftriaxone
in

Rifampicin

Erythromycin

Nafcillin

Azithromycin

Lincosamide

DIS: Doxycycline

CELL MEMBRANE ⊖

◆ **Polymixins:** Polymixin B and E (Colistin)

Colistin : Rx of NDM-I B lactamase (Carbapenem also resistant)

Produced by superbugs (Ex : Pseudomonas, Klebsiella, Acinetobacter)

◆ **Lipopeptide** (.....)

- DOC for VRSA and is also used for VRE
- MOA : binds to and depolarize the membrane producing cidal effect, Inactivated by surfactant
- SE : myopathy and neuropathy

ANTIMETABOLITES (DRUGS AFFECTING FOLATE METABOLISM)

➤ **Sulfonamides:** (Bacteriostatic): Sulfadiazine, Sulfamethizole, Sulfamethoxazole, Sulfadoxine

MOA: Sulfonamides are competitive antagonists of PABA, They inhibit the enzyme folate synthase aka dihydropteroate synthase

Uses:

- Silver sulfadiazine and Mafenide, Sulacetamide are topical creams in burn patients
- Sulfadoxine + Pyrimethamine is used in the Rx of malaria
- Sulfadiazine + Pyrimethamine is used in Rx of toxoplasma
- Cotrimoxazole (Trimethoprim+Sulfamethoxazole) in a ratio of 1:5 is DOC for Pneumocystis Jiroveci, Nocardiosis, Burkholderia. Its also used in Cystitis, Prostatitis and carriers of typhoid

(In Preg. :

Maternal toxoplasmosis diagnosed →

Step 1: Confirm fetal infection

Not confirmed → Spiramycin (safe any time, especially <18weeks)

Confirmed →

If <18 weeks Spiramycin (avoid pyrimethamine, teratogenic)

If >18 weeks →

Pyrimethamine + Sulfadiazine + Folic acid

- Sulfasalazine metabolite known as 5-ASA has anti-inflammatory property so its used in ulcerative colitis,
- Sulfacetamide is used for ocular infections,
- Sulfisoxazole for UTI

Pharmacokinetics

- These are highly protein bound drugs metabolized in the liver by acetylation.
- These drugs are not effective in the presence of pus, as pus contains large amount of PABA

Side effects:

- Skin rash and hypersensitivity reactions are common side effects
- They produce hemolytic anemia in patients with G6PD deficiency, SLE in slow acetylators
- Crystalluria
- Kernicterus in new born so C/I in neonates and pregnancy
- **Trimethoprim:** Its a bacteriostatic drug $\ominus$ DHF reductase of bacteria
- It used in Rx of prostatitis and UTI. It can cause megaloblastic anemia

PROTEIN SYNTHESIS $\ominus$

- ◆ Tetracyclines: Gram +ve and -ve
- MOA- they are analogs of t-RNA & cause competitive reversible inhibition at A site by binding to 30s sub-unit of ribosome.
- Tetracyclines are bacteriostatic

- **Ex:** Tetra/Demeclo/Doxy/Mino/Sare/Erava/Omada (cycline)
- Their absorption is impaired by food and also with Ca⁺, iron, milk etc hence given empty stomach
- Resistance to Tetracyclines develops due to formation of efflux pump

Uses: LGV, Lyme's dis, Leprosy, Atypical pneumonia, Anthrax, Rickettsia, Granuloma inguinale, Entamoeba, Chlamydia, Cholera, Brucella, Malaria (in patients travelling < 6 weeks), Urethritis (Mycoplasma hominis), and Px of Plague and (Large Corona Beer Makes U Pee)

- Tigecycline (glycylcycline) is broad spectrum also used in NDM - β - Lactamase, MRSA, VRSA, Sarecycline for acne

Side effects:

- Nausea, Vomitting is the MC side effect
- All are nephrotoxic except Tigecycline, Doxycycline, Minocycline which are safe in renal failure
- C/I in pregnancy, produce fetal tooth enamel dysplasia & retardation of bone growth
- Fanconi's syndrome by expired drugs. Demeclocycline & doxycycline can cause photo sensitivity
- Demeclocycline can cause diabetes Insipidus

◆ Aminoglycosides: Gram -ve

- MOA- misreading of new t-RNA at A site of 30s ribosome
- Ex: Strepto/kana/tobra/neo/spectino/capreo (mycin)

SAFE IN PREGNANCY

Pencillins
Azithromycin/ (Macrolides)
Cephalosporins, clindamycin
Metronidazole
Amoxillin+Clavulanate
Nitrofurantoin

genta/netil (micin) amikacin

- Aminoglycosides are bacteriocidal & show concentration dependent killing. They have poor oral absorption
- Resistance is acquired by enzymatic drug inactivation

Uses:

- Streptomycin, Gentamicin is the DOC in Plague, and Tularemia. Its also add on DOC in Pseudomonas and Enterococci
- Streptomycin, Amikacin, Capreomycin, Kanamycin (SACK) are used as 2ndline TB drugs
- Gentamicin, amikacin, are effective against gram -ve bacteria especially pseudomonas.
- Oral neomycin is given for gut sterilization in hepatic encephalopathy. The current DOC for the same is Rifaximin
- Spectinomycin for resistant Gonorrhea, Paromomycin for intestinal amoebiasis & kala azar

Side effects:

- Ototoxicity (mostly irreversible): Kanamycin, Amikacin, Neomycin cause hearing loss, Streptomycin and Gentamicin are vestibulotoxic. Netilmicin is least Ototoxic
- Nephrotoxicity (Reversible) : Least with Streptomycin, Max with Neomycin
- Neuromuscular toxicity: Potentiates neuromuscular blockades so these drugs are avoided in Myasthenia Gravis. Max with Neomycin and least with tobramycin

- ◆ **Chloramphenicol** - Binds to 50s ribosomes acts by inhibiting peptidyl transferase

Uses: Typhoid fever, H influenza, and severe Rickettsial infection

- Chloramphenicol is a broad-spectrum antibiotic but has limited use d/t side effects like Bone marrow suppression and Gray baby syndrome
- It is mainly bacteriostatic, It undergoes enterohepatic circulation, it is metabolized by glucoronidation
- Resistance to Chloramphenicol is by formation of inactivating enzyme acetyl transferase

- ◆ **Macrolides:** Gram +ve, Gram -ve

- MOA- inhibit translocation of t-RNA by binding to 50s ribosomes.
Ex: Spira/Clarithro/Azithro/Roxithro/Erythro/ (mycin)
Ketolide : Nafithro/Telithro mycin

- Resistance is seen by mutation of ribosome or production of ribosomal protective protein, Drug efflux or enzymatic breakdown

- ◆ Uses - Px of Rheumatic fever (penicillin allergy), Whooping cough, Atypical pneumonia (Mycoplasma), Legionella, Chancroid, Corynebacterium, Campylobacter, Cholera and Chlamydia in Pregnancy (alternative Furazolidone in cholera)

- ◆ Clarithromycin - DOC for MAC infection along with Ethambutol, Spiramycin is DOC for toxoplasma (preg)

- **Side effects-** QT prolongation and diarrhea d/t motilin stimulation (so used in gastroparesis and paralytic ileus)
- They are enzyme $\ominus$ (Erythromycin increases plasma level of Theophylline) , Cholestatic jaundice (usually by estolate salt of erythromycin)

◆ Lincosamide (Clindamycin & Lincomycin)

- MOA acts by binding to 50s ribosome & has similar action to macrolides
- Used in anaerobic infections (bacteroides, prevotella), TSS & gas gangrene and also used in chloroquine resistant malaria and pneumocystis jirovecii infections
- Topical Clindamycin is used in the Rx of acne has good activity against propionibacterium responsible for acne
Isotretinoin also used for Rx of acne

➤ Side effects: Pseudomembranous colitis

◆ Oxazolidinones: Act by binding to 50s ribosomes & $\ominus$ translocation and resistance is seen d/t mutation of binding site on ribosome

- Linezolid is used in MRSA, VRSA, VRE & also used in MDR - TB
SE : Thrombocytopenia & neutropenia (which blood monitoring should be done) its also a MAO $\ominus$ (Serotonin syndrome). Optic neuritis, Peripheral neuropathy and lactic acidosis are seen on long term use

- Tedizolid for MRSA and VRSA, Sutezolid for TB

◆ Streptogramins (Quinu / Dalfo**pristin**)

- MOA- They are bacteriocidal & acts by binding to 50s ribosomes. They are always given together for synergistic action.
USES: MRSA, VRSA, VRE

NUCLEIC ACID ⊖

- ◆ **Fluroquinolones:** They inhibit DNA gyrase or Topoisomerase II in Gram -ve bacteria which breaks down DNA

Pharmacokinetics:

- Spar / Moxifloxacin are longest acting
- Enz interaction : Cipro (Enz⊖)
- Levo (Least) Max oral bioavailability

Uses:

- Norfloxacin & Nalidixic acid are used in UTI whereas Moxi & Trova are ineffective (Secreted in bile)
- Delaflox (Rx : MRSA), Nadiflo/Ozeno **xacin** (Rx : Skin infections)
- Levo, Moxi, Gemi, Gatifloxacin has a good activity against gram +ve, Gram -ve like Klebsiella and Atypicals like Chlamydia and Legionella so they are aka Respiratory quinolones
- Ciprofloxacin is used in, Pyelonephritis, Prostatitis, Px of meningitis (Pefloxacin also can be used in meningitis), Anthrax, Shigellosis, Traveller's diarrhea, Typhoid (PAST)
- 4th Generation like Moxifloxacin, Trovafloxacin has widest spectrum

Side effects:

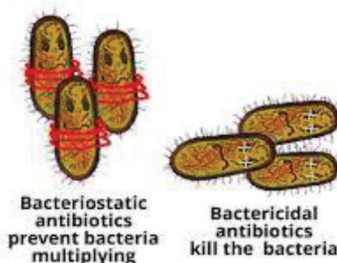
- Nausea Vomiting is most common side effect. Pseudomembranous colitis max with Ciprofloxacin
- Cartilage damage in children so C/I in children and pregnancy. Tendinitis is common in elderly
- QT prolongation, (Spar, Moxi) Photosensitivity (Pefloxacin)

- **Nitrofurantoin** causes DNA damage & is used as urinary antiseptic. It has no action against pseudomonas & proteus. SE: hemolysis in G6PD, neurotoxicity & phototoxicity
- Phenazopyridine is used as a urinary analgesic used to relieve dysuria & burning micturition
- **Nitroimidazoles:** Metro/Ti/Sec/Satranidazole (longest t $\frac{1}{2}$)
MOA: They get bioactivated to form reactive cytotoxic products that damage DNA
Uses: Metro is DOC for *Gardenerella*, Giardiasis, Entamoeba, Trichomoniasis, Anerobes (*Bacteroides*)
SE: Metro causes metallic taste, disulfiram reaction
- **Fidoxamicin** - RNA polymerase $\ominus$, DOC for Pseudomembranous colitis
- **Rifampicin** - RNA polymerase $\ominus$ used in TB & Leprosy
- Rifampicin is an enzyme inducer & its derivatives are Rifapentine (longest acting) and Rifamycin

Side effects:

- Rash is MC followed by Nausea, Vomitting, Reddish secretions
- Respiratory syndrome (Flu like symptoms), and its avoided in purpura
- Rifampicin causes OCP failure, Warfarin failure, Anti-retroviral drugs failure

Bacteriostatic	Bactericidal
Erythromycin (macrolides) Clindamycin Sulfonamides Trimethoprim Tetracyclines Chloramphenicol	Vancomycin Penicillins Fluoroquinolones Aminoglycosides Cephalosporins Carbapenems Rifampicin



ANTI - TUBERCULAR DRUGS (ATT)

Px - INH +/- Rifampicin

Rx - Intensive phase (HRZE x 2 months), Continuous phase (HRE x 4 months)

Isoniazid (H)	Rifampicin (R)	Pyranzinamide (Z)	Ethambutol (E)
⊖ mycolic acid	⊖ RNA polymerase	⊖ fatty acid synthesis	⊖ arabinosyl transferase
Hepatotoxic (Z>H>R)	Safe in Renal failure	Hepatotoxic (most)	C/I in Renal failure
Kat-G, Kas A, InhA gene mut	Rpo-B gene mutation	Pnc-a gene mutation	Emb gene mutation
Peripheral neuropathy, Anemia, (sideroblastic), SLE Gynecomastia	Red-orange secretions, stains contact lens, flu symptoms	Hyperuricemia (Z>E)	Optic neuritis, Red-green color blindness

CSF penetration: H>Z>R. Isoniazid, Ethambutol, Cycloserine is associated with Psychosis

Rifabutin (Pseudojaundice, Uveitis, Arthralgia), Ethionamide (Hypothyroidism), Thioacetazone (Steven Johnson Syndrome)

Rx: BPaLM (Bedaquilline, Pretomanid, Linezolid, Moxifloxacin)

o MDR-TB (multi): Resistance to H+R

o Pre-XDR: MDR-TB + Fluoroquinolone

o XDR-TB (Extensively): Pre-XDR + Bedaquilline/Linezolid

Use : Resistant TB.

Group A	Group B	Group C
Use all 3 drugs	Use 1 or both drugs	Used when group A or B cannot be Used
- Levo/Moxp floxacin - Bedaquiline - Linezolid	- Clofazimine - Cycloserine	- Meropenem - Amikacin/Streptomycin - Delamanid - Pyrazinamide - Ethambutol - Ethionamide - P Aminosalicylic acid

Latest drugs:

Bedaquiline - ATP synthase inhibitor, SE: QT prolongation

Dela/Preto~~manid~~ - ⊖ mycolic acid synthesis, SE: QT prolongation

- ✦ TB+HIV: Start ATT first followed by HAART later, else There is risk of IRIS (immune reconstitution inflammatory syndrome). Change Rifampicin to Rifabutin, Avoid Thioacetazone

Rifampicin is an enzyme inducer, it interacts with OCP, Warfarin, HIV drugs (Double the dose of dolutegravir - 50mg B.D)

CD 4 count	When to start ART
<50 CELLS/MM ³	Within 2 weeks of starting ATT (early ART improves survival)
> 50 cells/mm ³	After 8 weeks of ATT (to reduce IRIS risk)
TB meningitis	Delay ART by 4-8 weeks, as IRIS can worsen CNS inflammation

LEPROSY RX

Tuberculoid leprosy: Dapsone + Rifampicin + Clofazimine for 6 months

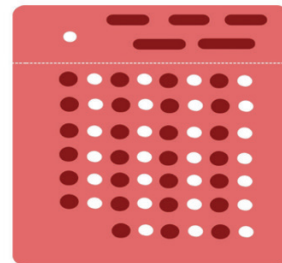
Lepromatous leprosy: Dapsone + Rifampicin + Clofazimine for 12 months

Lepra reactions: Steroids

Clofazimine SE: Hyperpigmentation, Ichthyosis



Day 1:	Dapsone(100mg x 1) Rifampicin (300 mgx2) Clofazimine (100 mgx3)
Day 2:	Dapsone (100 mg x1)
Onwards	Clofazimine (50 mg x1)



HAART (HIGHLY ACTIVE ANTI RETROVIRAL TREATMENT)

❑ **Entry ⊖:** Enfuvirtide (Gp41 ⊖), Fostemsavir (Gp120 ⊖)

❑ **RTI (Reverse transcriptase inhibitors)**

✓ **NRTI (Nucleoside) :** Zidovudine, Lamivudine, Emtricitabine

SE: BMS, LDS, Peripheral neuropathy, Lacticacidosis, Myopathy

✓ **NNRTI (Non-Nucleoside) :** Nevirapine, Efavirenz

✓ **NtRTI (Nucleotide) :** Tenofovir

❑ **Integrase ⊖ :** Ral / Dolu ()

❑ **Protease ⊖:** Tipra / Rito / Lopi / Daru ()

SE: LDS (by all), Renal stone (in dinavir)

◆ **Pre exposure Prophylaxis (PrEp) :** Tenofovir + Emtricitabine /
Carbo tegravir (long acting), Lenacapavir

- ◆ Post exposure Prophylaxis (PEP) : started within 2-72 hours after exposure and given for 28 days
 - ✦ Tenofovir+ Lamivudine+Dolutegravir / PI
- ◆ Mother to child transmission: Rx given to baby
 - ✦ If mother on ART:
 - ✦ If mother not on ART:
 - a. W/o Breast feed : Nevirapine + Zidovudine
 - b. With Breast feed : Nevirapine + Zidovudine
- ❑ Treatment :

Age > 10 Years : Tenofovir (TDF)+Lamivudine (3TC)/Emtricitabine + Dolutegravir (DTG)

or

- ❑ Children : Abacavir + Lamivudine + Dolutegravir / Protease ⊖

Tenofovir, Emtricitabine, Lamivudine has action against both HIV and Hep B

RSV : Ribavirin (Inhalational), Palivizumab, Nirsevimab

Herpes : A/Valacyclovir
In resistant cases : Foscarnet

Influenza : Osetla / zana **mivir**

Hep B : Tenofovir alafenamide > Entacavir (Doc), Adefovir, Lamivudine

Hep C : NS3/4 A (Protease -) Grazo/Sim/Parite/Tela**previr**

NS5 A Velpat/Pibren/Ombit/Ledip/Elb/Daclat**asvir**

NS5 B (RNA Polymerase) : Dasa/Sofos/Becla**buvir**

Active space

ANTI - MALARIALS

❑ Prophylaxis:

Before travel
(2 doses)



After going back
(4 weeks)

❑ Short term: Long :
(100 mg O.D) [250 mg]

❑ Rx

Species	Regular	Pregnant
Vivax	Chloroquine x 3days	Chloroquine
Falciparum	ACT (oral) x 3days	Quinine, ACT in 2nd & 3rd trimester
Mixed	ACT (oral) x 3days	Quinine, ACT in 2nd & 3rd trimester

Severe/Complicated malaria: Artesunate followed by oral ACT

ACT (Artemisinin combo therapy): Artemisinin drugs (Artemether/
Artesunate)

Rx

Artesunate 200mg
Sulphadoxine 500mg
Pyrimethamine 25mg

Rx

Artemether 20/60/80mg
Lumefantrine 120/360/480mg

Chloroquine: It's the drug hving largest Vd. It leads to heme accumulation
toxic to Parasite

Se : Bull's eye maculopathy, QT prolongation

Primaquine : Hypnozoite-cidal (14 days), Gametocyte-cidal (Single dose)

Vivax (0.25 mg / kg x 14 days), Falciparum (0.75 mg/kg single dose on day 2)

C/I in pregnancy, infants, G6PD

- DOC for Trematodes : Praziquantel except Fasciola (Triclabendazole)
- DOC for Cestodes : Praziquantel except Hydatid disease & Neurocysticercosis (Albendazole + Steroids)
- Doc for Nematodes : Albendazole wuchereria Bancrofti, Brugia malayi (Filariasis - Diethylcarbamazine), Strongyloides & Onchocera (ivermectin)
- Scabies (Caused by Sarcoptes scabiei mites) :
DOC Topical Permethrin (In preg also), Oral Ivermectin

ANTI - FUNGALS

1. Cell membrane ⊖

- a. Squalene epoxidase ⊖: Terbinafine, Butenafine, Naftifine.....
- b. 14- alpha demethylase ⊖:.....
- c. Polyene: Amphotericin B (AMB), Nystatin

2. Cell wall ⊖: Echinocandins (Caspofungin, Isavuconazole)

(β-1, 3 - D glucan synthase)

3. Protein Synthesis ⊖: Terbinafine

4. **DNA $\ominus$** : 5-Flucytosine

5. **Microtubule $\ominus$** : Griseofulvin

Azoles:

- **Fluconazole**: DOC for Candida & Cryptococcal meningitis (TOC: AMB + Flucytosine)
- **Itraconazole**: DOC for Histoplasma, Sporothrix, Blastomyces
- **Voriconazole**: DOC for Invasive Aspergillosis
- **Posaconazole**: Mucormycosis, Invasive Aspergillosis
- **Sertaconazole**: Anti-inflammatory & Anti-pruritic action
- **Ketoconazole**: Not preferred. SE: Gynecomastia, block steroid synthesis
- **Osetaconazole+Ibrexafungerp**: Oral Rx for Vulvovaginal Candidiasis

Polyene:

- **AMP-B**: DOC Mucormycosis, Cryptococcal meningitis, Kala Azar
SE: Infusion related like Fever, Chills (MC), Nephrotoxicity (RTA, Hypokalemia) that can be prevented by NS loading / combining with Liposomes (L-AMB)

Allyl-amines:

- **Terbinafine** (oral/topical): DOC for Dermatophytes (drugs accumulate in keratin rich tissues like Skin, Nail and Hair)

Griseofulvin: DOC for Tinea Capitis

Anti-Fungal Nail Lacquers

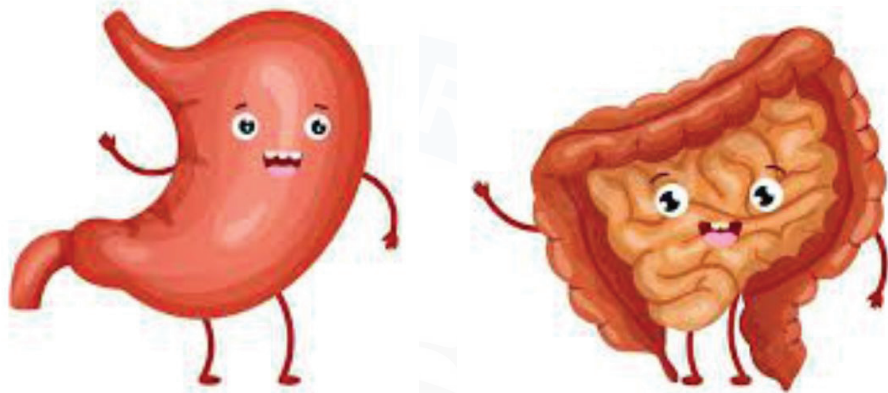
C: Ciclopirox olamine

E: Efinaconazole

A: Amorolfine

T: Terbinafine, Travoborole





Peptic ulcer disease:

Drugs to ↓ acid secretion: PPI (**prazole**) H2 blockers (**tidine**),
M1 blockers (**zepine**)

Drugs ↑ mucous secretion: Misoprostol, Enprostil, Rioprostil

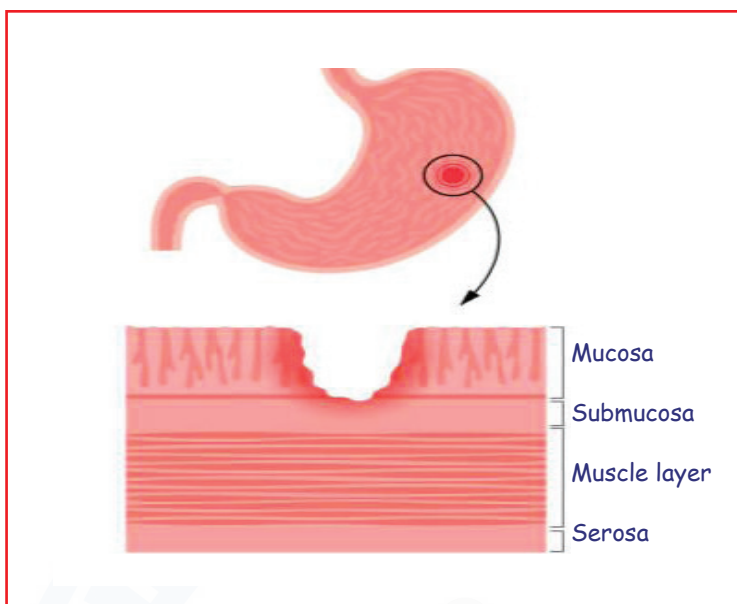
Drugs to neutralize acids: Antacids like Sodium bicarbonate,
 $\text{Al}(\text{OH})_3$, $\text{Mg}(\text{OH})_2$, CaCO_3

Drugs treating *H. pylori*: Metronidazole/Amoxicillin / Tetracycline
Clarithromycin, Ome/Lansoprazole/Vonoprazan, Bismuth

Triple regimen : 2 Antibiotics + 1 PPI

Quadruple regimen :

Ulcer coating drugs - Sucralfate, Bismuth



◆ PPI: Ome/Esome/Lanso/Rabe/Panto/Ila/Tenatoprazole

MOA: prodrugs (active form-sulfenamide) that irreversibly $\ominus$ $H^+ - K^+$ ATPase pump (parietal cells). Acid labile hence enteric coated

- PPI are DOC for PUD (any cause), Zollinger Ellison Syndrome, GERD, Stress ulcer
- Fastest: Rabe, Omeprazole, Long acting : Tenatoprazole
- Drug interactions: Least - Rabe, panto. Max - Omeprazole
- Most potent: Ila (5 mg) > Lansoprazole (15 mg)
- As PPI are enzyme $\ominus$, they $\downarrow$ activity of Clopidogrel, Tamoxifen. They have short $t_{1/2}$ (2 hrs) but long action (Hit & Run drugs)

SE: Headache, Diarrhea and abdominal pain.

$\downarrow$ B12, $\downarrow$ Ca (hip fractures), $\downarrow$ iron absorption,
 $\uparrow$ infections like Pseudomembranous colitis & pneumonia

◆ PCAB (Potassium competitive acid blockers): Vono/Reveprazan

Bind reversibly to K ions $\rightarrow$ blocks $H^+ - K^+$ ATPase pump. Faster onset than PPI

◆ H₂ antagonists: Cime / Rani / Niza / Roxa / Famotidine

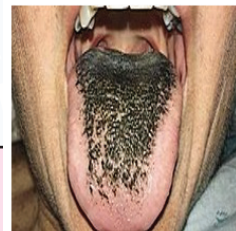
- These drugs competitively inhibit H₂ receptor on parietal cells
- Famotidine is most potent & longest H₂ blocker, Nizatidine has anti-cholinergic activity
- Cimetidine is least potent & is not preferred because it's an enzyme $\ominus$ ($\uparrow$ theophylline levels) crosses BBB, $\uparrow$ prolactin levels. Lafutidine has Nitric oxide activity $\rightarrow$ Mucus production $\rightarrow$ Healing ulcer

◆ Antacids:

- Antacids are weak bases that neutralizes the gastric acid but do not $\downarrow$ acid production
- Their major role is to provide prompt relief from ulcer pain
- Sodium bicarbonate (NaHCO_3) is fastest but not preferred long term because it releases CO_2 causing gastric distension & belching
- Antacids decrease absorption of acidic drugs like Tetracyclins
- Simethicone is anti-flatulence agent
- Aluminium hydroxide causes constipation whereas Mg salts produces diarrhea so combo is used
- SE: Milk-Alkali syndrome (Hypercalcemia, Renal insufficiency, Alkalosis) from excessive dose of Ca carbonate and Na bicarbonate

GASTROPROTECTIVE AGENTS

- ◆ Drugs ↑ protective factors: PGE1, PGE2 & PGI2 causes ↑ mucus and bicarbonate production, ↑ mucosal blood flow & also ↓ acid production
 - Misoprostol (PGE1 analog) is the most specific drug for prevention & Rx of NSAID - induced ulcer (DOC is PPI)
 - ◆ Ulcer coating drugs:
Sucralfate and colloidal bismuth
Sucralfate + Oxetacaine (Local Anesthetic)
 - Bismuth has additional anti H.pylori effects. SE: blackening of tongue
- Emetics: Apomorphine (Dopamine agonist), Ipecacuanha
 - They should not be used for kerosene & corrosive poisonings

**ANTI - EMETICS**

- ◆ Morning sickness:
- Doxylamine + vit B6 is the safest anti - emetic in pregnancy
- ◆ Motion sickness:
- Scopolamine/Hyoscine - used as transdermal patch as DOC for prophylaxis
- Antihistaminics - promethazine, cyclizine, diphenhydramine & meclizine can also be used for Px

◆ Chemotherapy & Radiotherapy induced Vomiting:

◆ **5HT₃ blockers:** Ondan (short t_{1/2})/Palono (most potent, long t_{1/2}), Grani/Dola/Ram^{setron}

- They are DOC for early-onset chemotherapy and radiation induced vomiting
- 5HT₃ blockers are preferred for post-operative vomiting & are also safe in pregnancy

◆ **Neurokinin-1 (NK-1) blockers:** Apre/Fosapre/Netu/Rola^{pitant} (longest t_{1/2})

- They are DOC for late-onset chemo & radiation induced vomiting

◆ **Steroids :** Dexamethasone

◆ **CB 1,2 ⊕ :** Dronabino (Synthetic tetrahydrocannabinol (THC))
Also used for anorexia in AIDS, cheonic pain (off Label)

PROKINETIC DRUGS

Uses : gastroparesis, GERD, post - op ileus.

- 5HT₄, CCK₁ & motilin receptors are stimulatory & D₂ is inhibitory

➤ **D₂ blockers:** Metoclopramide, Domperidone, Levosulpiride

- Metoclopramide - D₂ blocker, 5HT₃ blocker, 5HT₄ agonist. It crosses BBB so it can cause EPS and hyperprolactinemia

- Domperidone - lipid insoluble so no EPS. It's mainly used as antiemetic (↓ L-dopa vomiting w/o ↓ its efficacy)

➤ **5HT₄ agonist:** Cisa/Mosa/Renza/Ito/Prucalo^{pride}, Tegaserod

- Cisapride, Tegaserod has been withdrawn d/t QT prolongation & MI, stroke respectively

➤ **Motilin receptor agonist:** Erythromycin, Mitemcinal, Camcinal

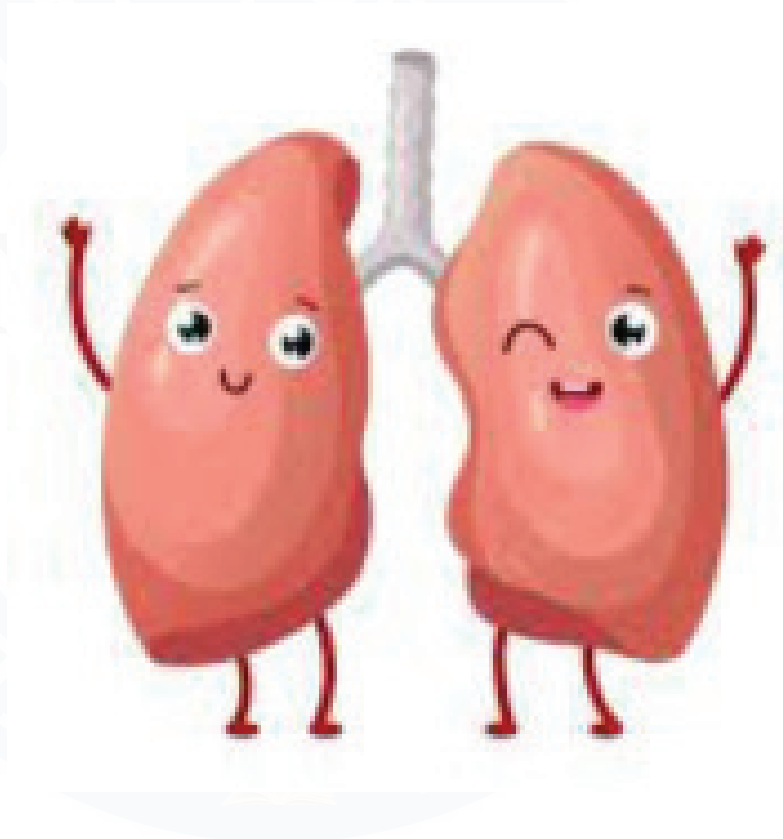
DIARRHEA

- **Opioids:** Loperamide (non-addicting), Diphenoxylate + atropine (to ↓ addiction)
- Loperamide (μ agonist) is DOC for non-secretory diarrhea like IBS. Opioids are C/I in infective diarrhea
- **Somatostatin analog** - Octreotide is DOC for secretory diarrhea like Carcinoid syndrome, VIPoma.
- Clonidine is indicated for chronic diarrhea with diabetes
Racecadotril : reduces hypersecretion of water & electrolytes in lumen
- Q. Ringer lactate (isotonic crystallised) - (ScaLP- Sodium, Chloride, Calcium, a, Lactate, Potassium)

CONSTIPATION

- ◆ **Bulk - forming:** Dietary fibre, Bran, Methyl cellulose, Psyllium (Metamucil)
- ◆ **Osmotic agents:**
 - ◆ **Saline laxatives** - Mg Citrate, MgSO₄, MgOH₂, Sodium phosphate
 - ◆ **Non-digestible sugars** (Lactulose, Mannitol, Sorbitol), Polyethylene glycol
- ◆ **Surfactant (Stool softner):** Docusate sodium, Calcium, Mineral oil, Liquid paraffin
- ◆ **Stimulants:** Bisacodyl, Cascara, Senna, Castor oil, Phenolphthalein, Sodium picosulfate
- ◆ **Fastest laxatives:** rectal stimulant or saline enemas / suppositories They work within minutes





Active space

Rx of cough

- ◆ **Rx of dry cough:** Antitussives (Codeine, Dextromethorphan, Noscapine)
- ◆ **Rx of wet cough:** Expectorants (Guaifenasin, KI) Mucolytics: Acetylcysteine, Bromhexine, Ambroxol

ANTI TUSSIVES

- Suppress cough by acting on CNS or inhibiting cough impulses in respiratory tract
- They should be used only for dry cough which include central acting drugs like Opioids (Codeine, Pholcodeine), Diphenhydramine, Dextromethorphan & Peripheral acting drugs like Cromolyn, Local anesthetics (Lidocaine, Bupivacaine), Moguistine
- Codeine is for mild to moderate cough. MC opioid side effect is constipation. Dextromethorphan is less potent opioid used for mild cough with less side effects
- Diphenhydramine is an antihistaminic. Sedation & anticholinergic side effects
- Mucolytics breaks mucus and thins it out and expectorants enhance expulsion

RX OF ASTHMA

◆ Bronchodilators: $\beta 2$ agonist, Anticholinergics, Methylxanthines

✓ Methylxanthines: Theophylline(oral), Aminophylline (iv)

MOA: $\ominus$ PDE (3 & 4), $\ominus$ adenosine receptors: B.D

$\oplus$ histone deacetylase-2 (HDAC-2): anti-inflammatory effect

SE: Headache, Arrhythmia, GI (NV) d/t PDE-4 $\ominus$

Diuresis, Arrhythmia, Seizures d/t Adenosine $\ominus$

(Mnemonic: HAGen DAS icecream) It's a drug with low TI (n:5-15 mg/L, toxicity: >20 mg/L)

✓ PDE-4 $\ominus$: Roflu, Cilo, Tofi **milast**

✓ PDE (3 & 4) $\ominus$: Ensifentrine

◆ Monoclonal antibodies:

✓ IgE $\ominus$: Omalizumab

✓ IL-5 $\ominus$: Mepo/Benra/Res **lizumab:**

✓ IL-4 13 $\ominus$: Dupilumab

✓ TSLP (thymic stromal lymphopoietin) $\ominus$: Tezepelumab

◆ Mast cell stabilizers: Sodium cromoglicate, Nedocromil, Ketotifen (has additional anti-histaminic action)

◆ Anti-leukotrienes:

LOX $\ominus$: Zileuton

LT receptor $\ominus$ (Monte / Zafirl **eukast,**)

PLA2 $\ominus$: Corticosteroids

✓ LT receptor $\ominus$ are associated with Churg-Strauss syndrome

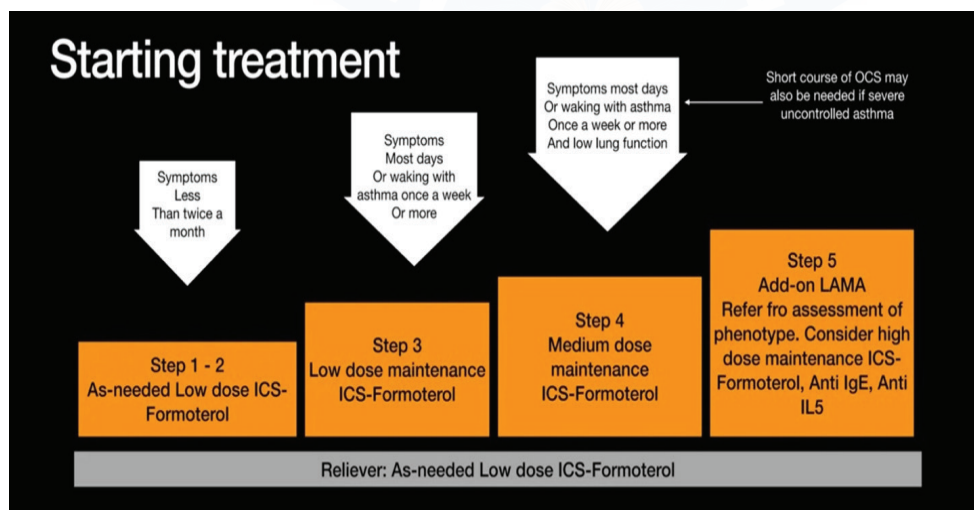
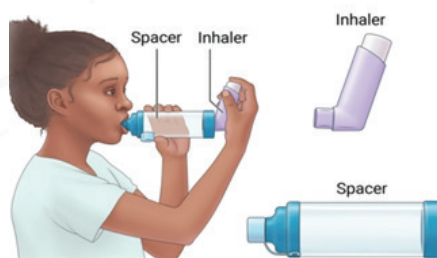
✓ **Steroids:**

- **ICS** (Fluticasone, Beclomethasone, Mometasone, Flunisolide, Budesonide, Ciclesonide)
- **Systemic** (Hydrocortisone, Methylprednisolone)
- **Oral** (Prednisolone, Methylprednisolone)

MOA: $\ominus$ PLA2, $\oplus$ histone deacetylase-2 (HDAC-2). They $\downarrow$ mucus secretion & bronchial hyperreactivity. They also $\uparrow$ sensitivity to β_2 agonist by $\uparrow$ β_2 receptors

Ciclesonide is metabolised in lungs & has less side effects hence k/a "soft steroid"

SE: Oropharyngeal candidiasis, Hoarseness of voice



Persistent Asthma (any type)	Inhaled corticosteroids (ICS)
Aspirin-Induced (AERD)	Leukotriene Receptor Antagonist (montelukast)
Exercise-Induced	SABA before exercise or daily LTRA
Allergic asthma	Omalizumab (anti IgE)
Eosinophilic asthma	Mepolizumab/Benralizumab/Dupilumab
Severe uncontrolled asthma	Add-on LAMA (tiotropium)

Rx of pulmonary fibrosis : Pirfenidone, Nintedanib

Rx of cystic fibrosis : Ivacaftor CFTR potentiator

(↑ C1 flow → hydrate airways, thin mucus and improve ciliary function)